

MAGNETOM

Troubleshooting Guide System

Gradient

Symphony, A Tim System

Document Version

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1.1 General

In this section, you will find the strategy and procedures for troubleshooting the gradient system

The main tool for gradient troubleshooting consists of error messages delivered from the gradient system and shown in the error log file of the service platform. A detailed description about the error is part of the error message. All message IDs are added in a table in this TSG. Each error message includes the error ID, message texts, explanations and actions. The gradient communication and status test is also available in the service software platform under "Test Tools".

To understand the basics of the gradient cabinet, please refer to the FUN (Gradient System).

Additionally required documents

FUN Chapter (Gradient System)

REP Chapter (Gradient)



Further troubleshooting informations e.g. error codes can be found in the Service Knowledge Base (SKB).

1.1.1 Overview of the gradient system:

System	Symphony TIM
option	Quantum
Gradient Cabinet	K2217 Cas. Quantum
Field Strength [mT/m]	30 mT/m
Software configuration	N4 VB12T 2000V/380A
Rise time [µsec]	300 µsec
Slew rate [mT/m/msec]	125T/m/s for Z 100 for X and Y
Field of View	500 mm
Board Ensemble	D11/112, D12, D14, D16 (CAN), D17, D 139, D180 (Backplane), D100, D110
Regulator Adjustment	Software-based tune-up step
Stimulation Monitor	SW stimulation in the AMC PCI-MON
Gradient coil	AS39T
Amplifier	3x power stage E6

System	Symphony TIM
option	Quantum
Gradient Cabinet	K2217 Cas. Quantum
Cooler	RCA
Cooling: Heat dissipation	30kW

1.1.2 Sources for errors in the gradient system

Compared to most of the other system components, the gradient system contains a number of assemblies that generate and process analog signals. These assemblies, all field replaceable units, (FRUs), form a closed current control circuit which makes it at times difficult to limit the source of error to a single FRU. Errors are usually looped across several assemblies. For example, the backplane D180 of the gradient small signal unit (GSSU) as well as all plug-in connections of the boards (PCB's) may be the source for an error.

Usually, if error messages are sent by the application software (ASW) to the AMC via the CAN-bus at the same time, the modulators will be blocked. The gradient current control is disabled and the Regulation error message is generated as a consequence.

Except for image quality errors, the following major types of errors are stored in the system error log (system/report). Several of them are also indicated via LEDs on the assemblies of the GSSU:

- **Dynamic errors:**

- These errors occur during measurements: They are either reproducible under the same conditions or they appear sporadically. In either case, the measurement is terminated. In most cases, the error message generated corresponds to the actual situation.

- **Static errors:**

- These errors messages are generated during GPA/KAS switch-on. Reset will not remove these errors. The source for error is an actual defect in the component or an error in the monitoring/transfer path (monitoring error).

- **Image quality errors:**

- Image quality depends on the error-free interactions between all system components; the most important of these are the RF and gradient systems. No specific error message will be generated when image quality is degraded (→ Image Quality - Test Strategy / Page 8).

1.1.3 Error handling - Error codes

Error messages are forwarded directly or via software-controlled multiplexer (on D14 K2217, Modulator PCB) to the CAN processor.

As they occur, all changes of the GPA/KAS status are forwarded from the CAN processor on D16 K2217 to the MPCU via a CAN telegram such as "Alarm", "General Error" or "Warning".

The type of telegram to be used is established in the application software of the GPA/KAS CAN unit. The definitions for the three types of telegrams are:

Alarm:	Fatal error or error status which will terminate a measurement in progress or prevent the start of a measurement. Whenever possible, the error will be indicated independent of the MPCU or MPCU software (e.g. red LEDs in the GSSU).
General Error:	Serious error or error status which will terminate a measurement in progress or prevent the start of a measurement.
Warning:	Error status which neither affects the measurement in progress nor prevents the start of a measurement. However, the image quality may be affected.



Further troubleshooting informations e.g. error codes can be found in the Service Knowledge Base (SKB).

1.1.3.1 Different STOP modes for the KAS:

MOS: All modulators will be inhibited via loadware by "SOFTSTOP", (i.e., freewheeling of power stage/gradient coil circuit, avoiding stimulation of the patient).

MOD OFF: All Modulators will inhibited by hardware.

POW OFF: The KAS will be switched into stand-by mode by loadware, i.e., transformer T1 off, power stage voltages are off.

TS OFF: The KAS will directly switch into stand-by mode, i.e., transformer T1 off, power stage voltages are off.

1.1.4 Test of CAN communication and GPA/KAS status

If you select "GPA/KAS Status" within the "Test Tools" platform of the service software, the following two quick tests will be performed:

- Test of the CAN communication between GSSU/D16 with AMC/cPCI CAN3.
- A listing of the GPA/KAS status will be displayed. Error conditions are marked with "!!!".

In case of errors, the displayed diagnosis will give you hints for additional troubleshooting.

1.1.5 Error reset and warm restart of GPA/KAS

The latched error signals (displayed by error LEDs on the GSSU) can be reset via the "Reset" button S3 on D17 K2217. Permanent errors are excepted.

Pressing the "CAN reset" button S404 on D16 initiates a warm restart, i.e. "Power on" sequence of the GPA/KAS. Static errors will then be logged and displayed again.

1.2 Image Quality - Test Strategy

1.2.1 General

The stability of the MR system is critical to the quality of MR imaging. Stability Check and CalcAr are routines used in the tune-up and quality assurance to monitor the stability of the MR imaging system. If the results of either or both of these routines are greater than a specific tolerance, the source of the instability can be found by investigating the various components that make up the MR imaging system. The service engineer would use the Gradient Tests (DAC, X,-Y,-Z-Axis, Functional) in the Test Tools service software platform when there is an indication that the source of the problem, e.g. instability, is in the gradient system:

For instance, when the instability measured in the QA functions of the service software "Stability Check" or "CalcAr" is orientation dependent, it is indicative of instability due to gradient instability and not RF or a magnetic field instability.

Other QA-functions check whether system parameters are adjusted correctly, e.g. Regulator check/ ECC Check or Graddelay. If the results are not within tolerance, you have to perform the corresponding measurements/adjustments of the tune-up.

In case of **orientation-dependent RFdisturbances in the image**, which may be caused by the gradient system (switching frequency of the amplifiers), you should interchange the gradient cables from the KAS of the affected axis in order to locate the origin of the problem. If the disturbances disappear with swapped cables, the problem is probably located in the GPA/KAS. Otherwise the problem must be connected with the gradient/body-coil system.

1.2.2 Summary of gradient dependent IQ - Parameters

IQ Parameter	Influence Factor	Available Tests
Spatial Resolution Contrast (Symptoms are edge rings and edge blurring)	Non-linear amplitude of gradient pulses Sequence timing related to eddy current fields	Check linearity of gradient pulses (Gradient Loops); eddy current compensation (ECC)
Slice Thickness Slice Profile Slice Position Slice Warp	Eddy currents Linearity of gradient field	ECC Depending on gradient coil design; no improvement possible
Artifacts such as smearing artifacts Ghosts: discrete/blurred	Phase encoding errors: Any kind of gradient instability; periodic unequal phase encoding steps SENS instability/nonlinearity	DAC: Data loop test Check gradient stability/linearity (Gradient loops) Regulator adjustment

Image Uniformity Spatial Linearity	Non-linear gradient magnetic fields	No improvement possible
Signal to Noise Ratio	Gradient system: Eddy currents; spikes generated by gradient switching	ECC, gradient delay; Spike check

1.2.3 Gradient System - Image Quality TSG Chart

Normally there are no specific error messages corresponding to an image quality problem. You have to specify the individual problem according to the "Symptom" column in the table below and perform the proposed test to localize the problem in the MR-system.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Smearing artifacts in phase encoding direc- tion: ■ discrete ghosts ■ blurred ghosts	a. Low phase encoding per- formance of the gradient system caused by instability	SENS instability and offset at low gradient amplitudes D10 DAC defective/wrong gradi- ent data from MPCU/GCTX Regulator oscillation	Perform the Gradient Tests; check the stability of the DAC/ actual value Replace D10 Check regulator adjustments	SeSo: TestTools TuneUp/QA Regulator Check

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
02. Edge blurring/rings artifacts in phase encoding direction	a. Nonlinearity of gradient pulse steps	Slew rate limitation of the gradient pulses at high amplitudes (may be caused e.g. by partially defective power stages)	Perform the Gradient Tests; check the linearity of the DAC/ actual value. Check function of power stages: The six green LEDs behind the holes in the front of the power stage must be on. Perform Gradient Performance Check/ Gradient Rise Time QA/Gradient Rise Time Check Perform Calcar and Calcar lem-shift if ghost artifacts exist.	SeSo: TestTools/ Gradient REP/ Gradient, Power stages TSG/ Gradient/ Procedures: Gradient Performance Check/ Gradient Rise Time QA: Regulator Check Troubleshooting GPA Current Sensors (→ Troubleshooting GPA Current Sensors / Page 26)
03. Streaks and/or dots in images	a. RF noise caused by rapid gradient switching	Magnet room is not RF-tight	Check RF-room shielding, filter plate, RF-filters	

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
04. Low signal to noise ratio and/or spikes	a. Gradient system: Gradient/Bodycoil system	Eddy currents; spikes caused by mechanical vibrations during gradient switching	Perform - Graddelay - ECC Check - Spike Check *)	SeSo: TuneUp/ QA
05. Non-uniformity of images (phantom image)	a. Gradient system	Eddy currents	Perform ECC Check	
06. Image spatially distorted	a. Adjustment of gradient system	Gradient sensitivity misadjusted	Perform Gradsens	
07. Low contrast/resolution of the image: Edge rings and blurring	a. Gradient system Nonlinear amplitude switching of gradient pulses	Wrong gradient data D10 DAC	Check gradient data transfer: Perform AMC Tests and Gradient Tests (check results of DAC/functional Test) Perform the gradient linearity tests DAC/actual value	SeSo: TestTools AMC Gradient
	b. Gradient system	Eddy currents	ECC/Check	SeSo: TuneUp/ QA
08. Wrong slice position and/or warped slice	a. Shim condition of magnet	Distorted magnetic field	Check the shim condition of magnet/ Reshim the magnet	SeSo: TuneUp Shim/ Phantom Shim
	b. Gradient coil design	Non-linear gradient field	No improvement	
09. Broadening of slice thickness/profile	a. Gradient system	Eddy currents	ECC/Check	SeSo: Tuneup/ QA

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
10. Insufficient fat saturation	a. Base magnetic field	Magnetic field inhomogeneities due to bad shim (possible reasons may be: magnet quench, magnetic materials in the magnet bore)	Perform Shim Check. Perform Phantom Shim. Reshim the magnet.	SeSo/ QA: Shim Check TuneUp: Phantom Shim
	b. Gradient offsets	Gradient offset currents have changed (linear gradients)	Perform Shim Check. Perform Phantom Shim. Reshim the magnet.	
	c. Eddy currents	Uncompensated eddy currents/ cross term eddy currents	Perform ECC Check/ ECC adjustment Perform CTC Check/ CTC adjustment	SeSo/ QA: ECC Check CTC Check TuneUp: ECC/CTC
*) Note: TRUFI and HASTE sequences and special sequences used e.g. for EPI or BOLD imaging generate spikes, which cannot be verified by the normal QA "Spike check". In these cases, use the procedure "advanced spike" in the QA expert mode for spike troubleshooting. The "advanced spike" procedure should be repeated at least three times - the system has to be free of spikes.				

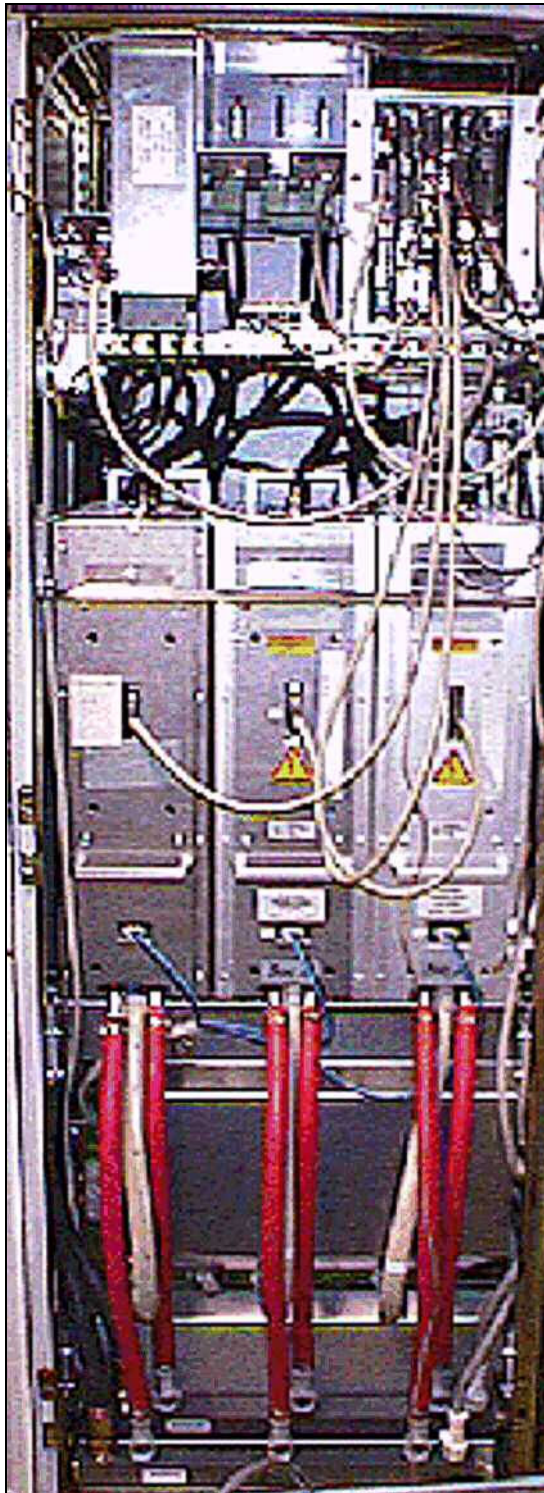
1.3 Summary of FRU's and tests for K2217

FRU's K2217 (→ Procedures / Page 16)	Test procedures
GSSU	
DAC board D11(2) (57 25 507/ 73 69 973) (→ DAC D11(2) Test Procedures / Page 20)	a. Check jumpers and switch position b. Offset measurement/adjustment c. Gradient data test loop /TestTools/AMC
Regulator board D12 (57 26 349 K2217) (→ Regulator D12 Test Procedure / Page 22)	a. Default regulator setting by DIP switches b. Jumper settings for maximum current, current regulation mode and current direction dependent switching c. SENS offset measurement/adjustment d. Regulator Adjust/Check
Modulator board D14 (57 26 364) (→ Modulator D14 Test Procedure / Page 24)	a. Check of slew rate jumper settings
Service/CAN board D16 (57 25 515 K2217) (→ Service/CAN D16 Test Procedures / Page 29)	a. Check of communication to the main power supply. b. CAN bus communication (Test Tools/Gradient/GPA status).
Loop/CAN board D139 (→ Loop/CAN Board D139 Test Procedures / Page 30)	a. Check jumper X3: 1- 3 b. TestTools/Gradient c. TestTools/RCA d. TestTools/AMC
I/O board D17 (57 26 430 K2217) (→ I/O D17/Power supply Test Procedures / Page 31)	a. Switch setting of S1 and S2 b. Jumper settings X9 c. Supply voltage measurement
GPA cabinet	
Small signal power supply E4A (30 59 482) (→ Small Signal Power Supply Unit E4A Procedures / Page 33)	a. Measurement/Adjustment

FRU's K2217 (→ Procedures / Page 16)	Test procedures
Current sensor assembly (SENS) (57 25 655 K2217) (→ Regulator D12 Test Procedure / Page 22)	a. Offset adjustment of D12 b. TuneUp/ Regulator adjustment c. TuneUp/ Phantom Shim d. Tune/Up Grad Delay e. TuneUp/ Grad Sens f. QA/ ECC Check g. QA/ Image Orientation h. QA/ CalcAr i. Calcar/ Calcar Iemshift (→ Troubleshooting GPA Current Sensors / Page 26)
GPA cabinet	
Main power supply boards:	
Power stage (59 38 142)	a. TuneUp/ Regulator adjustment b. TuneUp/ Phantom Shim c. TuneUp/ Grad Delay d. TuneUp/ Grad Sense e. QA/ Image Orientation
	f. (→ Gradient Performance Check / Page 39)
Blowers (30 63 476) (→ GPA Fans Test Procedure / Page 44)	Function check
Soft start unit D31 (57 26 414) (→ D31 Soft Start Unit Test Procedure / Page 34)	a. Check LEDs and connectors
Rectifier unit D32 (57 27 016) (→ D32 Rectifier Unit Test Procedure / Page 37)	a. Check connector b. Measurement of supply voltages for power stages

2.1 GPA KAS K2217

Fig. 1: Front side of GPA_059 cabinet



Components:

(from left to right) temperature sensor (above E4A), power supply unit E4A & fan line filter SENS (in the back) GSSU & fan cabling, FOC's

Power stages X,Y,Z-axis

NF filters (in the back)

5 fans

Transformer T1 (in the back), main contactor K2

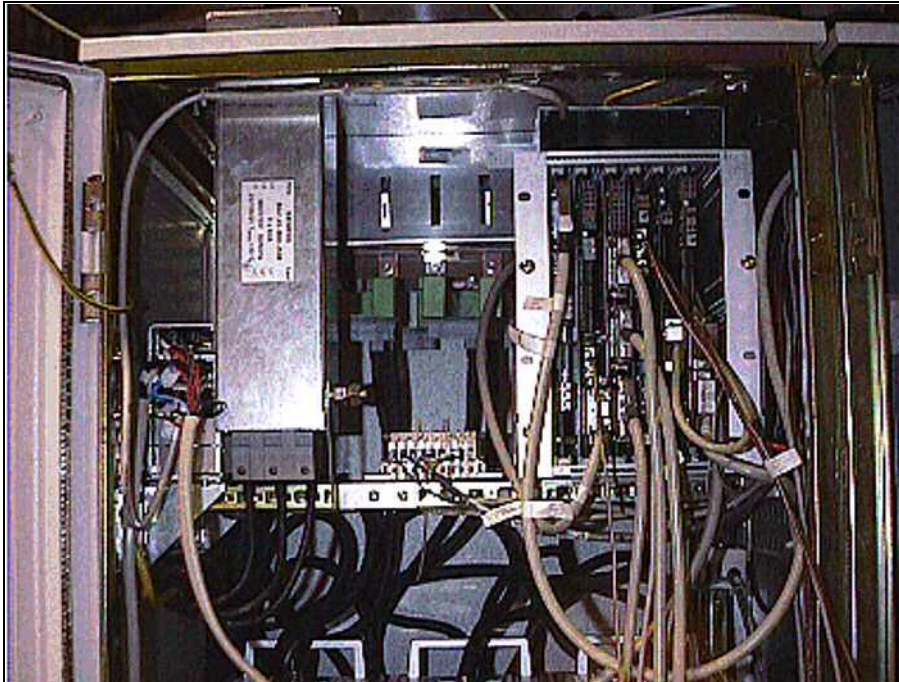
D31, Soft start unit

D32, Rectifier board

Cooling water distributor

2.1.1 FRU's in the upper part of the GPA Cabinet

Fig. 2: Upper part of GPA cabinet



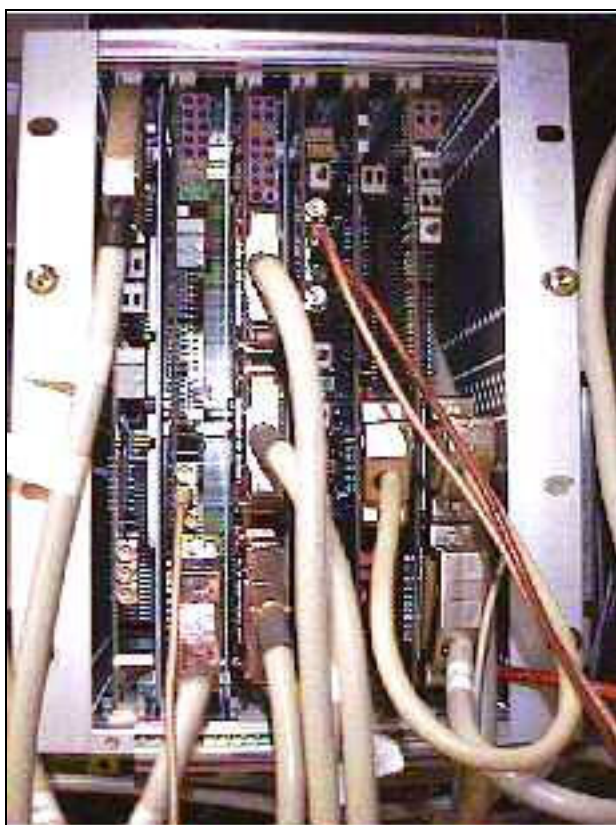
Components from left to right

- Power supply unit E4A
- Line input filter
- Connector block X100
- GSSU
- In the background: Current sensor assembly

Position of PCBs:

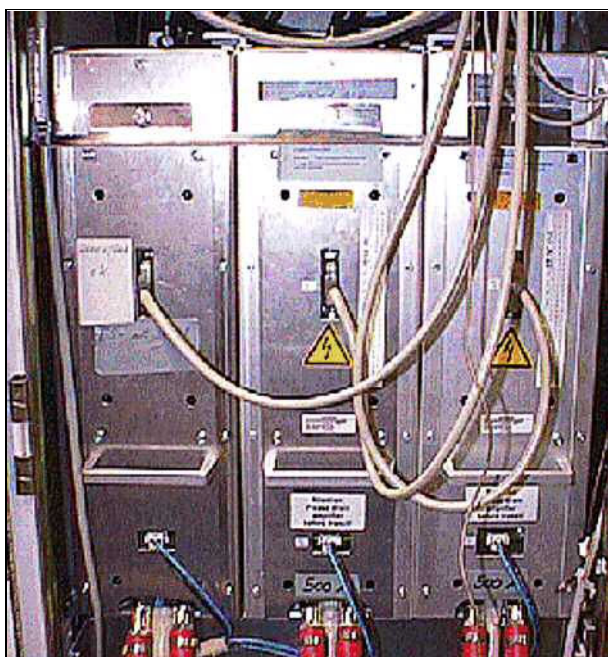
D11(2) - D12 - D14 - D16 - D139 - D17

Fig. 3: GSSU of GPA K2217



2.1.2 FRUs in the middle part of the GPA Cabinet

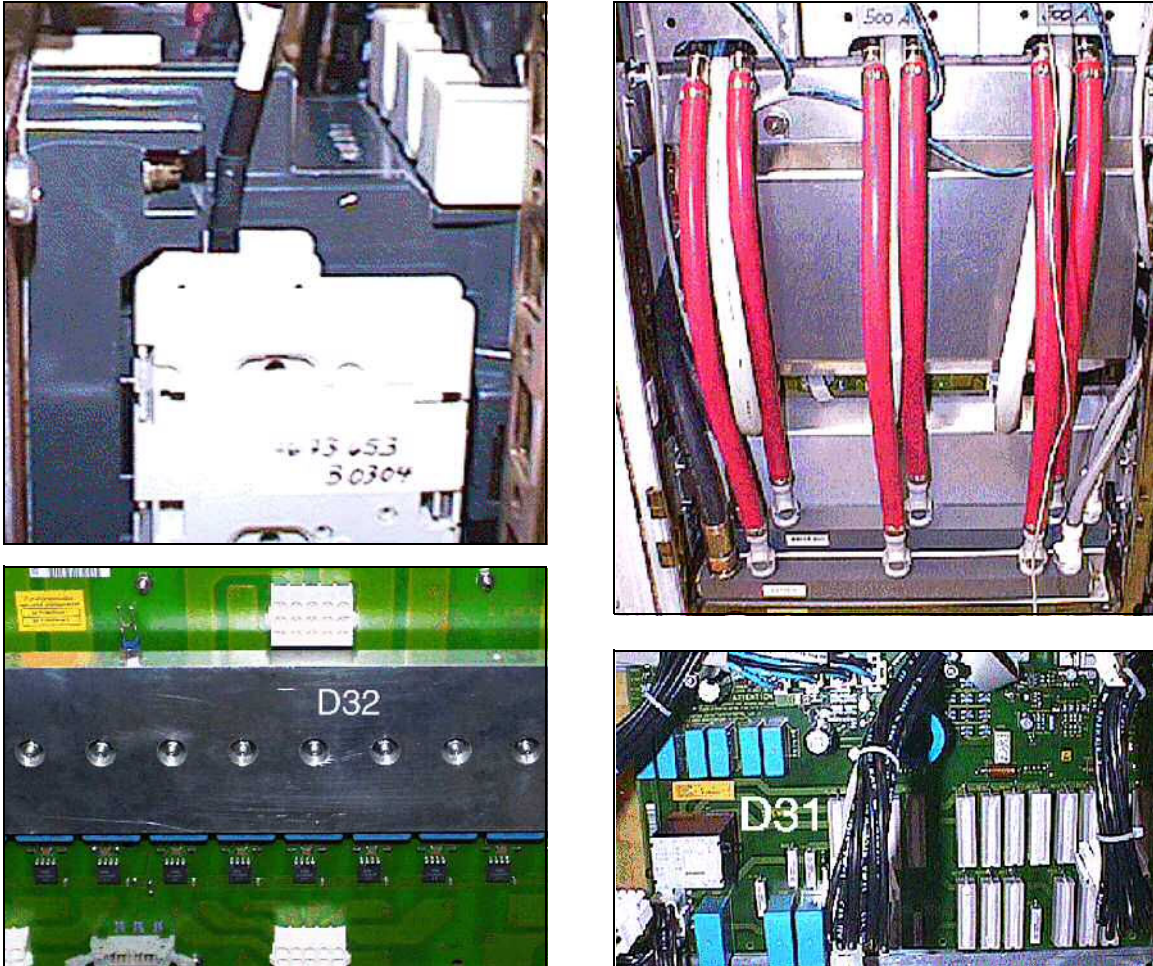
Fig. 4: Middle part of GPA cabinet



X,Y,Z-Power stage with cabling and water pipes

2.1.3 FRUs in the lower part of the GPA Cabinet

Fig. 5: Lower part of GPA cabinet (with D32 partially shown left and D31 right and main contactor K2)



Fans below the power stages

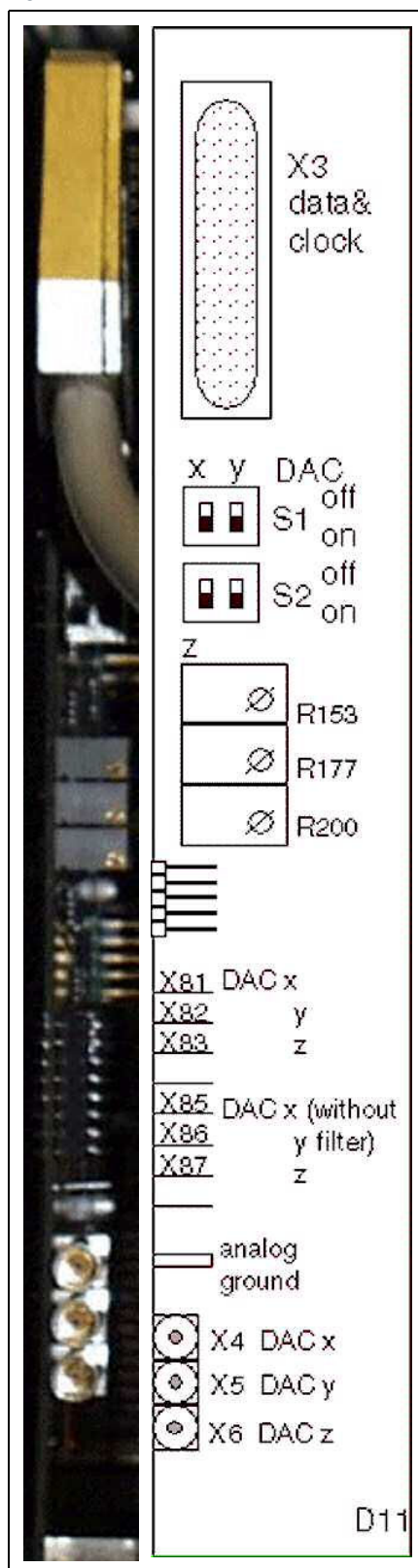
Behind cover plates:

- Rectifier board D32 in front
- Soft start board D31 behind D32
- Main contactor K2
- 25A fuses (15fuses, blow slow)

Water distributor with water pipes

2.1.4 DAC D11(2) Test Procedures

Fig. 6: Front of DAC board D11



Descriptions:

X3:

20 bit parallel gradient data (18 bit used) and 4MHz clock signal from AMC

Gradient data: D11/X3 - AMC/D5/X11

S1/S2:

The DAC output can be switched on/off for every axis; S2/2 is not used.

Potentiometers for offset adjustment

DAC output = nominal value (0-> +/-10V) 10 V corresponds to the maximum output current.

a. Check jumpers and switch positions:

Jumper X16: 2-3 (sign of gradient data, 2-Complement)

Rotary switch S3: Position 3

Rotary switch S4: Position 6 (this setting results in a clock signal FSYNC = 20 kHz)

b. Offset measurement/adjustment:

Prerequisite is the regulator adjustment (Tune-up).

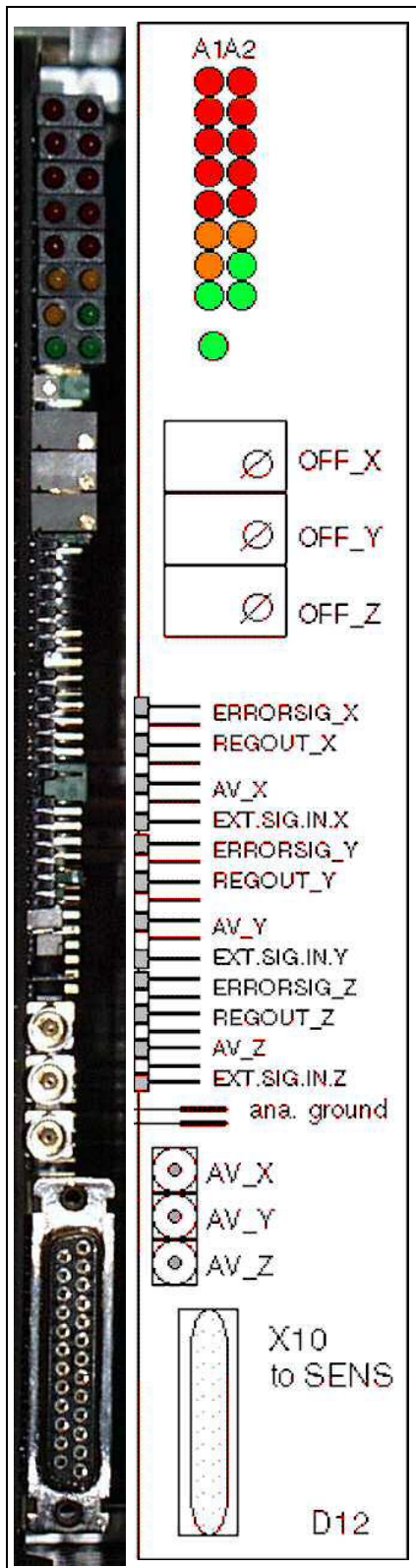
- Gradient cabinet has been on for at least 10 minutes.
- Set switches S1/2 to up position ("DAC output zero")
- The corresponding modulator must be on: S1/2 on D4 in down position.
- Connect a DVM to the desired actual value output X11,12,13 on D12. Use the QLA-BNC service cable.
- Check the offset: Offset < +/- 0.05 mV
- Use the corresponding R153, 177, 200 on DAC-board D11 to adjust the offset for the X, Y and Z axes if necessary.

c. Gradient data test loop AMC/D5/X11 - GSSU/D11/X3:

Use the test "Gradient/ DAC/Functional" of the TestTools platform to confirm the correct gradient data transmission. between AMC and GSSU/D11; (a prerequisite for this test is the correct function of the AMC components, refer to AMC tests).

2.1.5 Regulator D12 Test Procedure

Fig. 7: Front of Regulator D12



Descriptions:

LEDs (A1 left / A2 right):

V130: IMAX_X / IMAX_Y V131: IMAX_Z / CAV_ERR_X

V132: CAV_ERR_Y / CAV_ERR_Z

V133: DUCY_X / DUCY_Y

V134: DUCY_Z / OHMLOSS

V135: LOOP_ERR_X / LOOP_ERR_Y

V136: LOOP_ERR_Z / REG_ON_X

V137: REG_ON_Y / REG_ON_Z

V159: current flow

Offset pots

X: R 155

Y: R 189

Z: R 223

Test points (refer to Function Description)

X11,12,13: Actual current value X,Y,Z
QLA (1V=50A)

X10 SENS connection

a. Default jumper setting by DIP switches (regulator parameters)

DIP-switch	Pos1	Pos2	Pos3	Pos4	Pos5	Pos6	Pos7	Pos8
S1,S2,S4 S5,S7,S8	On	Off	Off	Off	Off	Off	Off	Off
S3,S6,S9	On	On	On	Off	Off	Off	Off	Off

b. Jumper settings:

Jumper	Setting	Function
X16, X18, X20	1-2 2-3	Nominal value X,Y,Z from D11 Nominal value externally
X17, X19, X21	1-2 2-3	Current regulation normally Control from X67,77,87
X22/X23	1-2/1-2	Processor controls DAC gain
	2-3/2-3	DAC maximum range $\Leftrightarrow$ 505A
	open/2-3	DAC maximum range $\Leftrightarrow$ 435A
	2-3/open	DAC maximum range $\Leftrightarrow$ 375A
	without jumper	DAC maximum range $\Leftrightarrow$ 300A
X24	1-2	Switching depending on current direction
	2-3 do not use this position !	Switching not dependent on current direction

c. SENS offset measurement/adjustment:

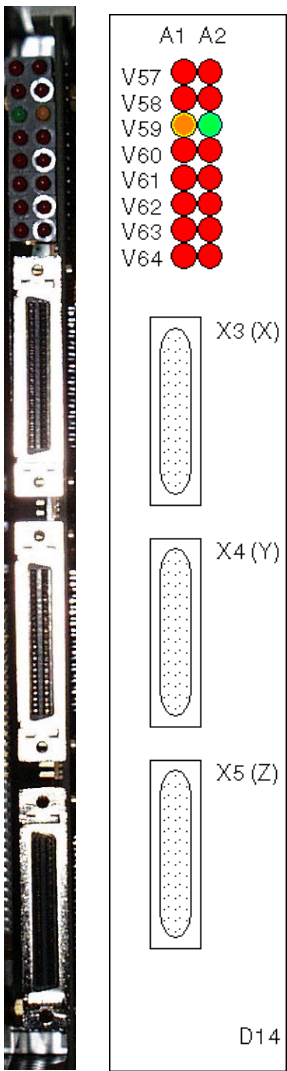
- Gradient cabinet has been on for at least 10 minutes and the regulators are adjusted (see Regulator Adjust/Check below)
- Connect the DVM to X11,12,13 ("actual value") on the regulator board. Use the QLA-BNC service cable.
- Set the switch, corresponding to the axis, on the I/O board D17 in up position ("Modulator off").
- Check the offset: Offset < +/- 0.05 mV; use offset pots to adjust the offset if necessary.

d. Regulator Adjust/Check:

In NUMARIS 4 the regulators are set via the Service Software function **TuneUp/Regulator Adjust**. The **QA/Regulator Check** function quickly checks the regulator data set.

2.1.6 Modulator D14 Test Procedure

Fig. 8: Front of Modulator D14



Descriptions:
Error LED's

X3,4,5: Connections to power stages X,Y,Z
Clock signals, control of drivers, power stage transistors and temperature supervision (50-pin connectors)
Note:
If one of these cable connections is defective, LED V62/A2 is on during state 1.

2.1.6.1 a. Meaning of LED's

LED	LED ON
V57/A1	N_CLOCK_ERR (error in 82 MHz clock)
V57/A2	M_MOD_ERR (modulator disabled)
V58/A1	P_15V_ERR (+15V error)
V58/A2	N_15V_ERR (-15V error)
V59/A1	SYNC_ERR_CH (GPA not synchronous to system clock)
V59/A2	5V_ok
V60A1/2	Continuously on: Supervision chip not configured, change D14 (D142)

LED	LED ON		
V60/A1/A2	State 1: A1 ON / A2 OFF	State 2: A1 OFF / A2 ON	State 3: A1 ON / A2 ON
V61/A1	N_INT_INT Modulator disabled	N_Spare	MOD SPARE 1
V61/A2	OFF_X, N_MODON_X X-modulator disabled	OFF_Y, N_MODON_ Y-modulator disabled	OFF_Z, N_MODON_Z Z modulator disabled
V62/A1	EMMI Danger of stimulation	Output disabled FOC control disabled	MOD_SPARE 3
V62/A2	Pull protection (cables D14/2 to power stages)	MOD_DEF Modulator D14 defective	MOD_SPARE 2
V63/A1	Switching frequency X too high	Frequency Y	Frequency Z
V63/A2	TEMP X Overtemperature PS-X	TEMP Y Overtemperature PS-Y	TEMP Z Overtemperature PS-Z
V64/A1	OVV_X Overvoltage PS-Y	OVV_Y Over voltage PS-Y	OVV_Z Over voltage PS-Z
V64/A2	Driver voltage X Too low	Driver voltage Y Too low	Too low

After state 3, the LED's V60, A1/2 are off, all 8 LED's below are on.

2.1.6.2 b. Check of Jumper Settings:

Slew rate settings:

Jumper	Setting	Function
X9/X10	1-2/1-2:	Processor controls slew rate
	2-3/2-3	Maximum slew rate possible
	2-3/open	n/a
	open/2-3	Half slew rate (SR2)
	without jumper SR4 (includes SR2)	Quarter slew rate (SR4)

(Refer to the Function Description for further explanations).

2.1.7 Troubleshooting GPA Current Sensors

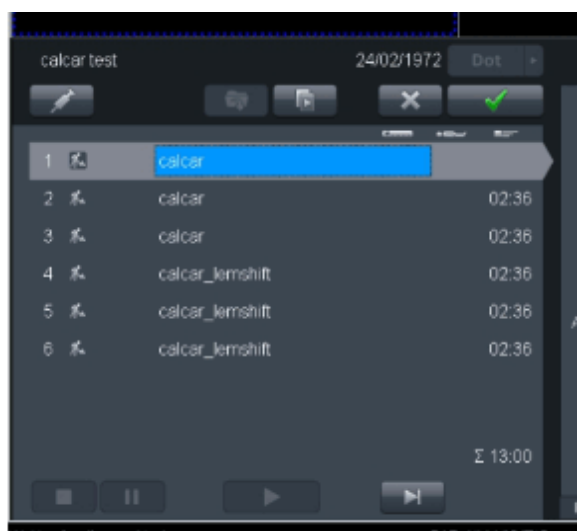
This document describes:

- How to find out defective (nonlinear) GPA current sensors.

2.1.7.1 Prerequisites

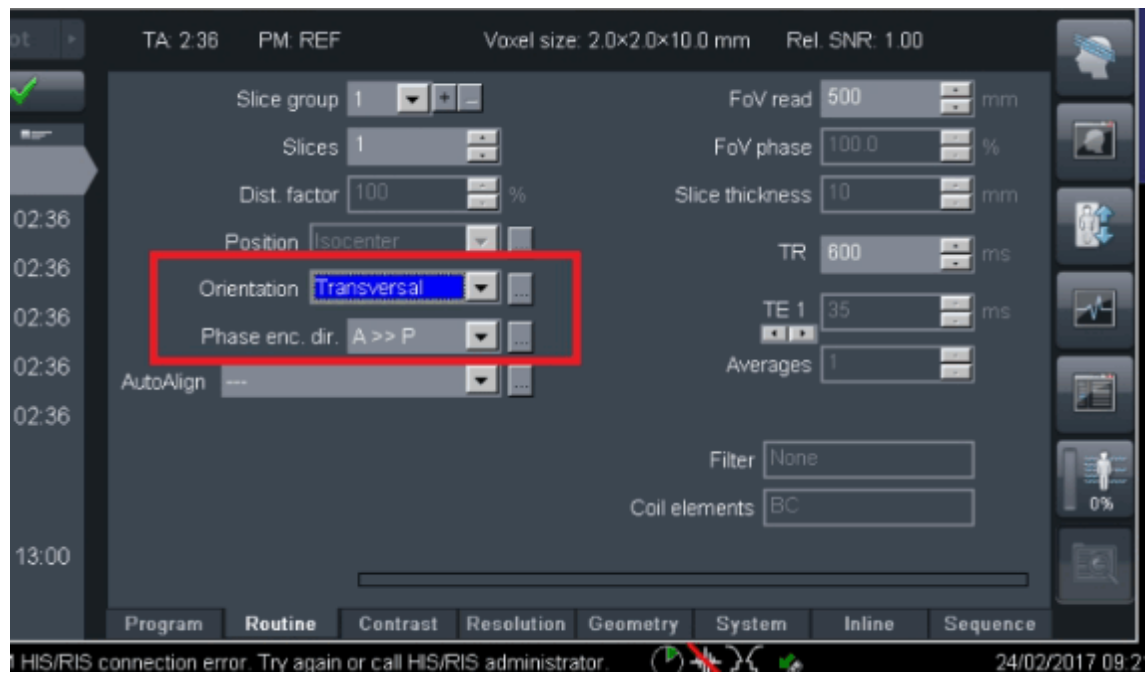
- Register a service patient
- Position a 240 mm spherical phantom (and loader for Tim systems)
- Navigate to the Siemens service sequences and copy 3 times calcar and 3 times calcar_lemshift

Fig. 9: Calcar test



- Calcar 1 – Orientation Transversal – Phase enc. Dir -> A-P
- Calcar 2 – Orientation Transversal – Phase enc. Dir -> R-L
- Calcar 3 – Orientation Coronal – Phase enc. Dir -> F-H
- Calcar_lemshift 1 – Orientation Transversal – Phase enc. Dir -> A-P
- Calcar_lemshift 2 – Orientation Transversal – Phase enc. Dir -> R-L
- Calcar_lemshift 3 – Orientation Coronal – Phase enc. Dir -> F-H

Fig. 10: Calcar test setup



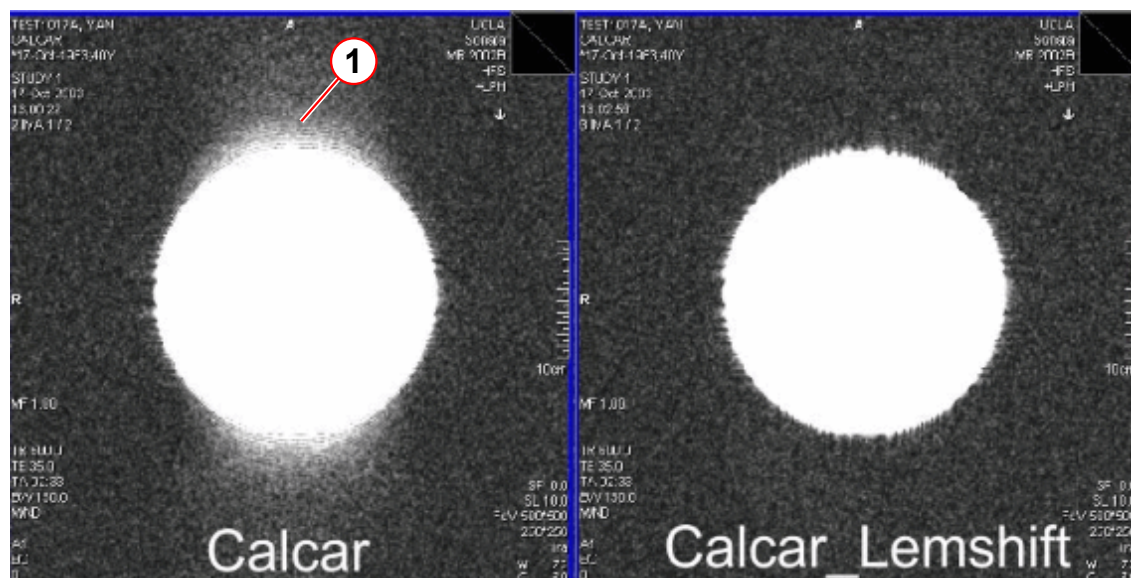
- Run the 6 sequences in a row
- Open the viewing task card with a 3x3 layout
- Load the images in such a way:

Calcar 1 (PE A-P) Y axis	Calcar 2 (PE R-L) X axis	Calcar 3 (PE F-H) Z Axis
Calcar_lemshift 1 (PE A-P) Y axis	Calcar_lemshift 2 (PE R-L) X axis	Calcar_lemshift 3 (PE F-H) Z axis
(empty)	(empty)	(empty)

- Compare the results of the two sequences for the same PE direction
- If calcar_lemshift is significantly better than calcar (less ghosting artifacts (→ Fig. 11 Page 28)), troubleshoot/ replace the corresponding current sensor of this axis.

Refer to REP Gradient.

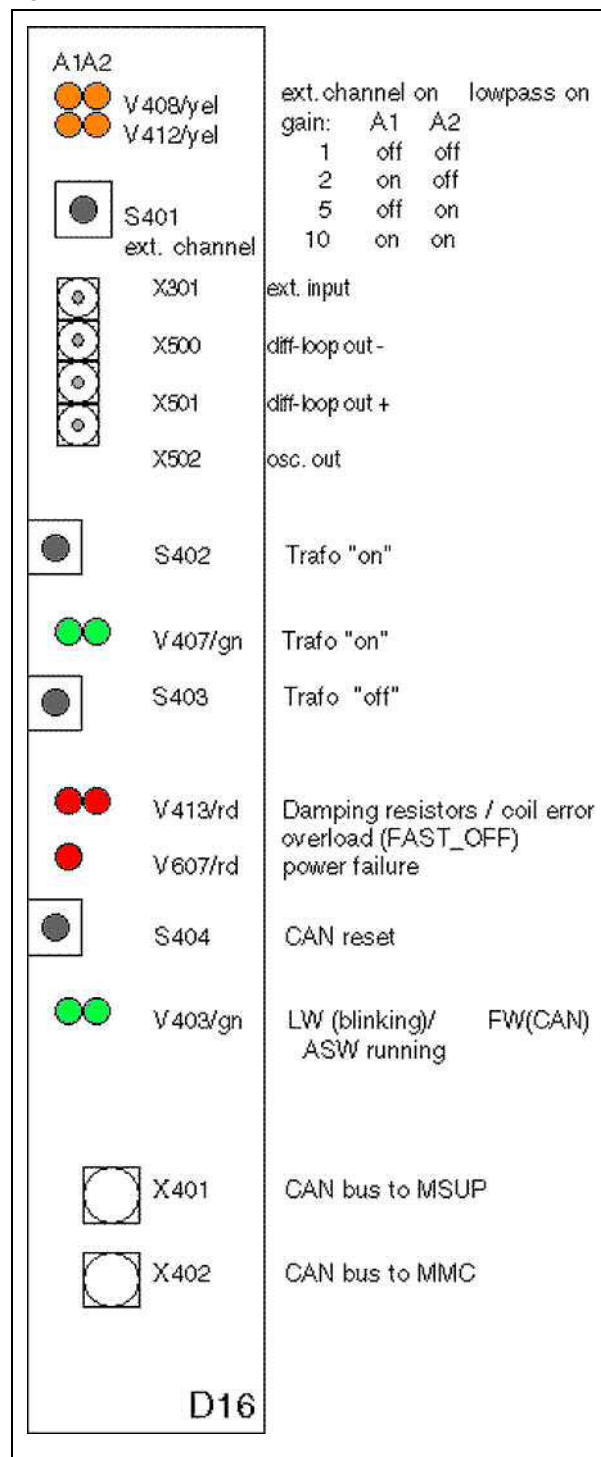
Fig. 11: Calcar test - defective current sensor



(1) Ghosting Artifacts close to phantom

2.1.8 Service/CAN D16 Test Procedures

Fig. 12: Front of Service/CAN board D16



Descriptions:

LEDs show conditions of external channel settings.

S1: Sets external channel, not used

X301 External signal input

X500, X501 not used

X502 Oscilloscope output

S402: Manual "ON"

V407: Transformer on/off

S403: Manual "OFF"

V413 left: Damping resistors overload
V413 right: Gradient coil error

S404: Warm start of GPA-processor

V403 left, blinking during normal conditions

Jumper X3: 1-2 (regulator D-portion enabled)

a. Check communication to main power supply unit:

The main power supply unit receives the switch-on signal via cable from assembly D17 (X1). Test by pressing S403 and then S402. Verify that the main power supply unit is switched to stand-by (LEDs "EN" and "RDY" off) and on again after approx. 15 seconds (switch-on signal is present at D17 (X1) and is transmitted to the main power supply unit via cable).

b. CAN bus communication/GPA status:

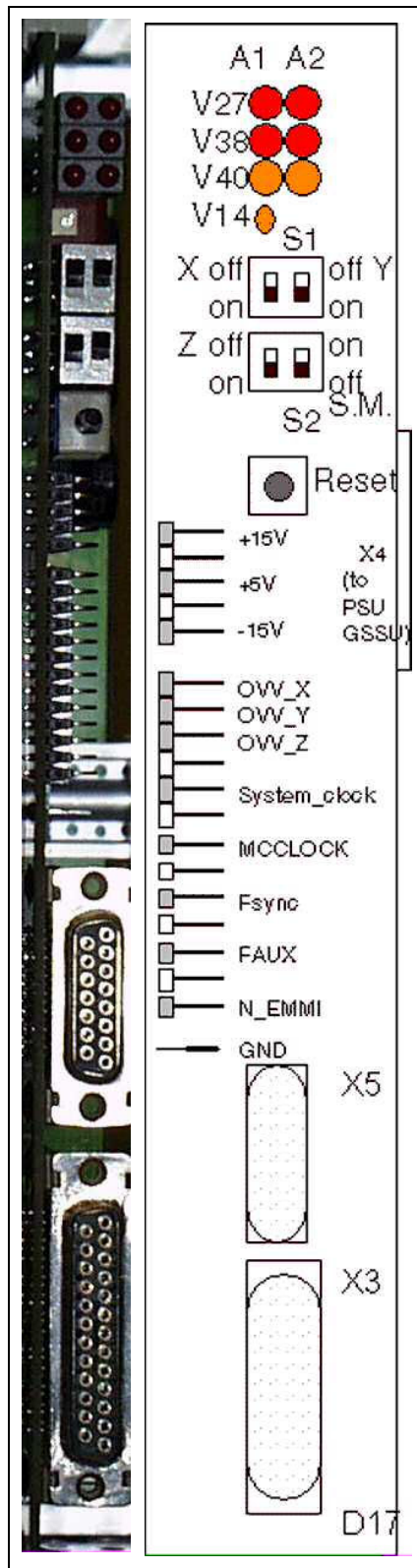
The present status (Status/warning/errors) of the GPA is displayed completely with the help of the Service Software function "**Gradient/Communication**" in the platform Test-Tools.

2.1.9 Loop/CAN Board D139 Test Procedures

1. Check jumper X3: 1-3.
2. Run TestTools/ Gradient.
3. Run TestTools/RCA.
4. Run TestTools/AMC.

2.1.10 I/O D17/Power supply Test Procedures

Fig. 13: Front of I/O board D17



Descriptions:

LEDs (A1 left / A2 right):

V27: Rectifier -Temp. / Transformer-Temp.

V38: Coilerr / Filterfan

V40: EMMI / DB_WDG_ERR (stim.mon.)

V14: Yellow LED: Service mode

Switches for modulator on/off and service mode on/off

Error Reset

Test pins:

+/-15V, +5V

Overvoltage X,Y,Z

System clock 4 MHz

Modulator clock 20 kHz

Synchron.clock 20 kHz

Auxiliary clock 50 kHz (not used)

X5 connector to D139/X51

X3 Connector to switch-on assembly D31

The I/O board D17 is the interface of the GSSU to the switch-on assembly D31 as well as the SENS (synchronization via three FOCs) and supervision assemblies such as D9 (coil supervision) and the stimulation monitor. Electrical signals are converted into FOC signals and vice versa.

Additionally, the D17 delivers the supply voltages (from E4A power supply unit) to the GSSU components. X4 on the board is the connector to the E4A.

The four switches at the front of the board allow an individual switch-off for every modulator and the selection of the service mode.

A push button allows the reset of an error situation.

In case of errors, this is an easy method for ruling out "monitoring errors":

The status of the error transmission path is tested by pulling the respective fiber optic cable.

"Light off" means that there is an error and/or one of the corresponding sensors is defective.

Check of pin positions jumpers D17/X9

Jumper	Link	Signal function
X9	11-12	Transformer temperature (no supervision)
	19-20	Filter fan (no supervision)

Connector D17/X3 (to soft-start unit D31)

Pin	Meaning	Pin	Meaning
1	K1/K2 switch on (p)	14	Message K2 closed
2	K1/K2 switch on (n)	15	+15V
3	Transformer overtemperature (p)	16	+15V
4	Transformer overtemperature (n)	17	+15V
5	Rectifier overtemp (p)	18	0V (GND)
6	Rectifier overtemp (n)	19	0V (GND)
		20	Shield (to GND)
		21	-15V
		22	-15V
		23	-15V
		24	Extr. power input (p)
		25	Extr. power input (n)

D17 FOCs

Connector	Meaning
U1,2,3	Synchronization clock to current sensors
U7	Temperature supervision gradient coil

2.1.11 Small Signal Power Supply Unit E4A Procedures

Fig. 14: Adjustment of supply voltage



Slot 1: +5.075 +/- 0.025 VDC. Measure the voltage on D17 (Fig. 106)

Slot 2: +16,5VDC, supply voltage for drivers at D21 on the power stages and fan on top of GSSU. Measure the voltage at the terminals of E4A (Fig. 107).

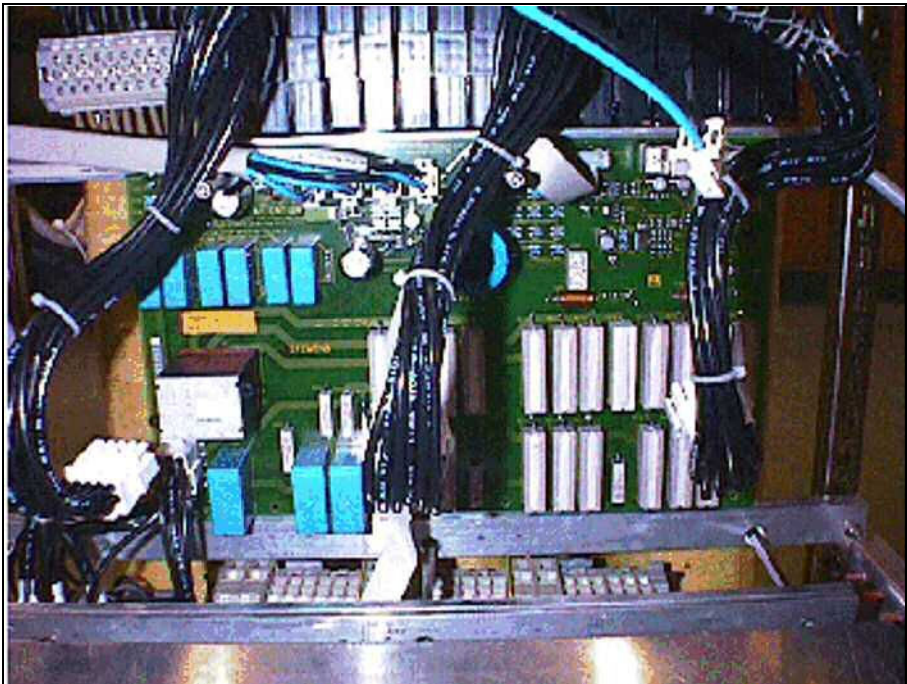
Slot 3: +/- 15.05 +/- 0.025VDC. Measure the voltage on D17 (Fig. 106).

Adjust the voltages with switched off modulators using the marked potentiometers (Fig. 107).

Slot	Terminal at E4A
Slot 1	A = 5V, B = GND, C = 5V (Sense), D = GND(Sense)
Slot 2	E = +Fan, F = GND Fan, G = +Driver supply, H = GND Driver supply
Slot 3	I = +15V, J = GND, K = GND, L = -15V

2.1.12 D31 Soft Start Unit Test Procedure

Fig. 15: D31 soft start unit (D32 rectifier board removed)



Six lighted LED's at left top of D31 indicate a closed motor protection switch F110/80A on top of ACC mains box, i.e. main voltage is present on D31.

**CAUTION**

Before switching on F110, switch off the GSSU with circuit breaker F6 in the ACC. Otherwise, severe damage to the D32 and the power stages may occur.

» Never switch on F110 (80A cutter switch-on top of ACC, if the GSSU is switched on.

Fig. 16: D31 Soft start unit - Voltage-indicating LEDs and driver supply connectors X11, 12, 13 (to PS) and X10 (from E4A)



Connector X1 to GSSU, D17/X3

Pin	Meaning	Pin	Meaning
1	K1/K2 switch-on (p)	14	Message K2 closed (n)
2	K1/K2 switch-on (n)	15	+15V

Pin	Meaning	Pin	Meaning
3	Transformer overtemperature (p)	16	+ !5V
4	Transformer overtemperature (n)	17	+ 15V
5	Rectifier overtemp (p)	18	0V (GND)
6	Rectifier overtemp (n)	19	0V (GND)
7	Overvoltage Z (p)	20	Shield (to GND)
8	Overvoltage Z (n)	21	-15V
9	Overvoltage Y (p)	22	-15V
10	Overvoltage Y (n)	23	-15V
11	Overvoltage X (p)	24	Overload (EPI) (p)
12	Overvoltage X(n)	25	Overload (EPI) (n)
13	Message K2 closed (p)	-	Shield

Connectors X11 to X13 and X10

Pin	Meaning	Pin	
1	+16,5V	2	0V

Flat cable connector X8 to D32/Rectifier board

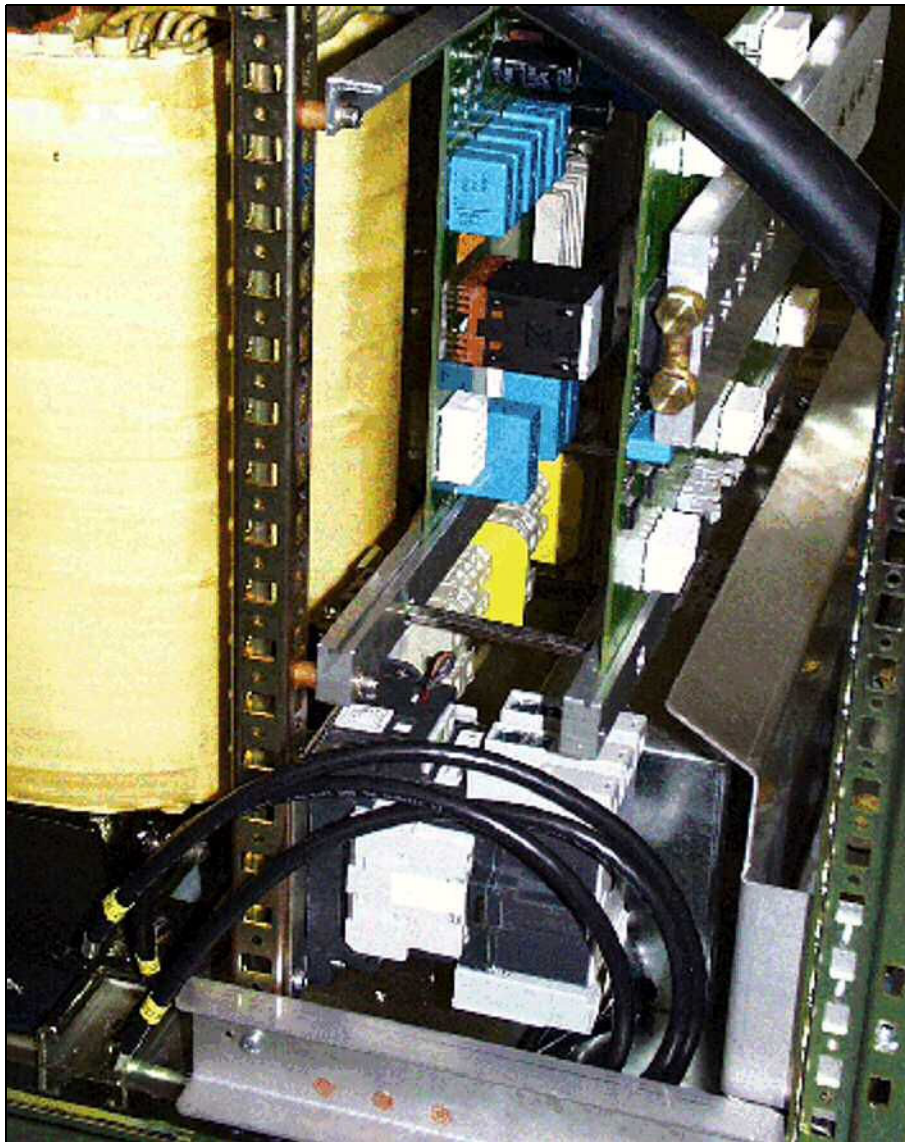
Pin	Meaning	Pin	Meaning
1	Connect. supervision	2	
3	+15V	4	+15V
5	0V (GND)	6	0V (GND)
7	PTC on rectifier(n)	8	PTC on rectifier (p)
9	Overvoltage X (p)	10	Overvoltage X(n)
11	Overvoltage Y (p)	12	Overvoltage Y (n)
13	Overvoltage Z (p)	14	Overvoltage Z (n)
15		16	Connect. supervis. (GND)

Connector X2 to K2 main contactor

Pin	Meaning	Pin	Meaning	Pin	Meaning
1	K2_A1_old	2	K2_A2	3	K2_A1_new
4	K2 meas. cont.	5	K2 meas. cont.	6	
7		8		9	

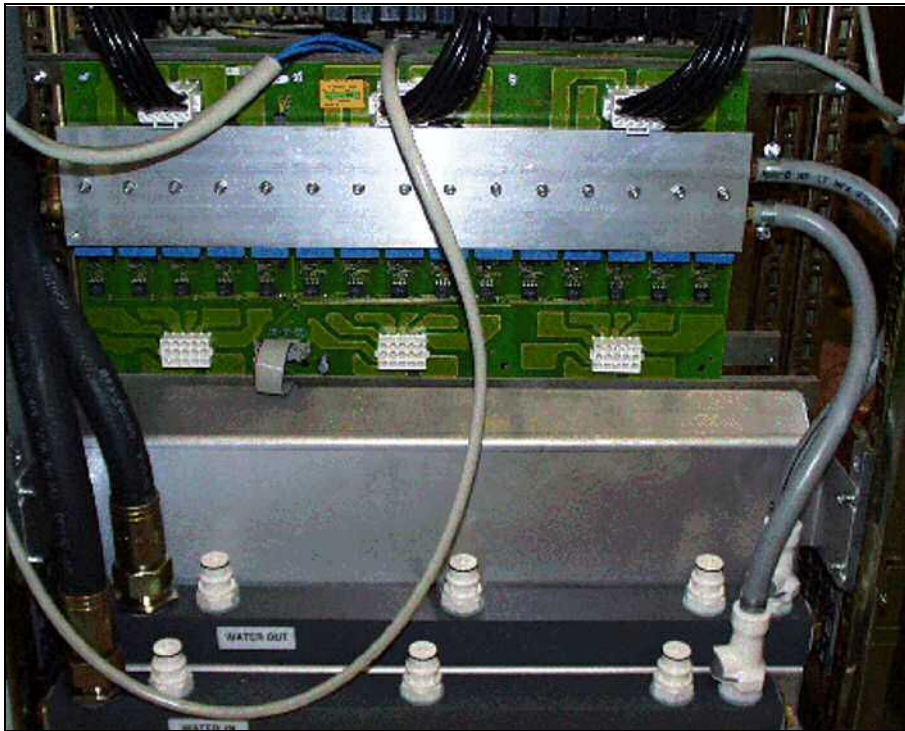
Pin	Meaning	Pin	Meaning	Pin	Meaning
10	Transformer L1	11	Transformer L2	12	Transformer L3
13	Line L1	14	Line L2	15	Line L3

Fig. 17: D31 Connector X2 - Main contactor K2



2.1.13 D32 Rectifier Unit Test Procedure

Fig. 18: Rectifier board, ACV power connectors from T1 and cooling water pipes plugged in



CAUTION

Before switching on F110, switch off the GSSU with circuit breaker F6 in the ACC. Otherwise, severe damage to the D32 and the power stages may occur.

- » Never switch on F110/80A cutter switch-on top of ACC, if the GSSU is switched on.

Connector X1 to D31/Softstart unit

Pin	Meaning	Pin	Meaning
1	Connect. supervision	2	
3	+15V	4	+15V
5	0V (GND)	6	0V (GND)
7	PTC on rectifier(n)	8	PTC on rectifier (p)
9	Overvoltage X (p)	10	Overvoltage X(n)
11	Overvoltage Y (p)	12	Overvoltage Y (n)
13	Overvoltage Z (p)	14	Overvoltage Z (n)
15		16	Connect. supervis. (GND)

Connector X11(X), X21(Y), X31(Z) ACV supply of rectifiers

Pin	Meaning X11 (X)	Meaning X21 (Y)	Meaning X31 (Z)
1	2U1	2W1	2V1
2,3	2U2	2W2	2V2
4	3V1	3U1	3W1
5,6	3V2	3U2	3W2
7	4W1	4V1	4U1
8,9	4W2	4V2	4U2
10	5U1	5W1	5V1
11,12	5U2	5W2	5V2
13	6V1	6U1	6W1
14,15	6V2	6U2	6W2

Measurement of supply voltages for power stages

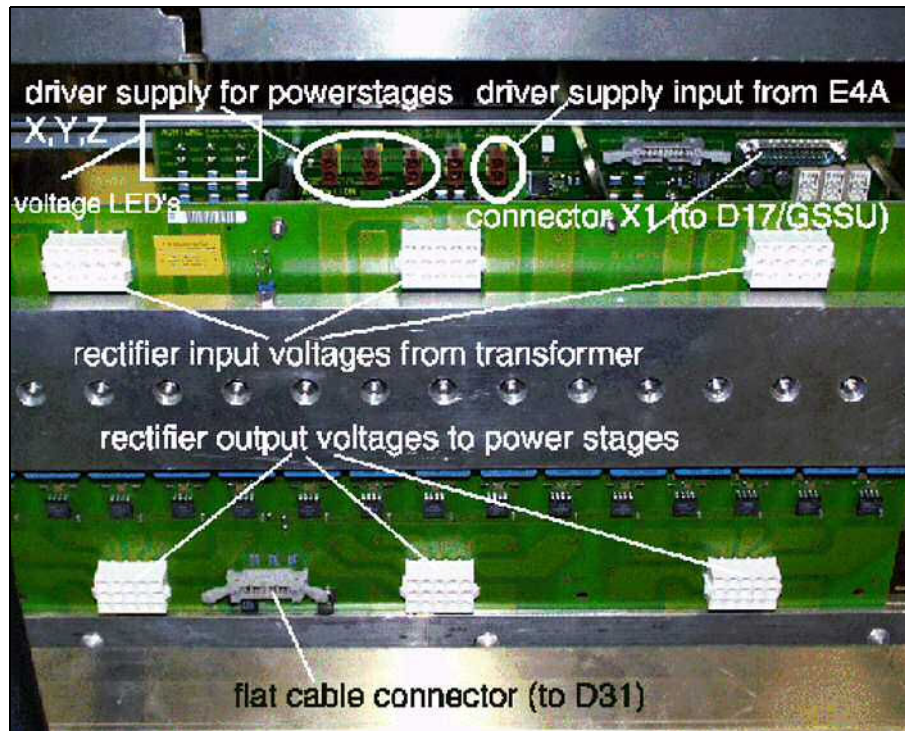
- Switch off K2/T1 with D16/S403.
- Switch off the cutter switch F110/3*80A on top of ACC.
- Pull one of the output connectors X12, X22, or X 32, (where you want to measure the output voltage).
- Connect a DVM (range 1000 VDC) with appropriate clips to the connector pins at D32 according to the table below.
- Switch off the modulators (D17 S1 and S2 left in upward position).
- Switch on the service mode (D17 S2 right in up position).
- Switch on F110/3*80A.
- Switch on K2/T1 with D16/S402.
- Read the output voltage at DVM: 400 +/- 2 VDC.
- Switch off K2/T1 with D16/S403.
- Switch off F110/3*80A on top of ACC.
- Remove the measurement equipment.
- Plug in the removed connector again.
- Switch on F110/3*80A on top of ACC.
- Switch on K2/T1 with D16/S402.
- Switch on the modulators (D17 S1 and S2 left in down position).
- Switch on the service mode (D17 S2 right in up position).

Connector X12(X), X22(Y), X32 (Z) DCV supply of power stages

Pin	Meaning	Pin	Meaning
1	DC voltage X,Y,Z_U1+	3	DC voltage X,Y,Z_U1-

Pin	Meaning	Pin	Meaning
4	DC voltage X,Y,Z_U2+	6	DC voltage X,Y,Z_U2-
7	DC voltage X,Y,Z_U3+	9	DC voltage X,Y,Z_U3-
10	DC voltage X,Y,Z_U4+	12	DC voltage X,Y,Z_U4-
13	DC voltage X,Y,Z_U5+	15	DC voltage X,Y,Z_U5-

Fig. 19: Plug positions on D32/D31



2.1.14 Gradient Performance Check

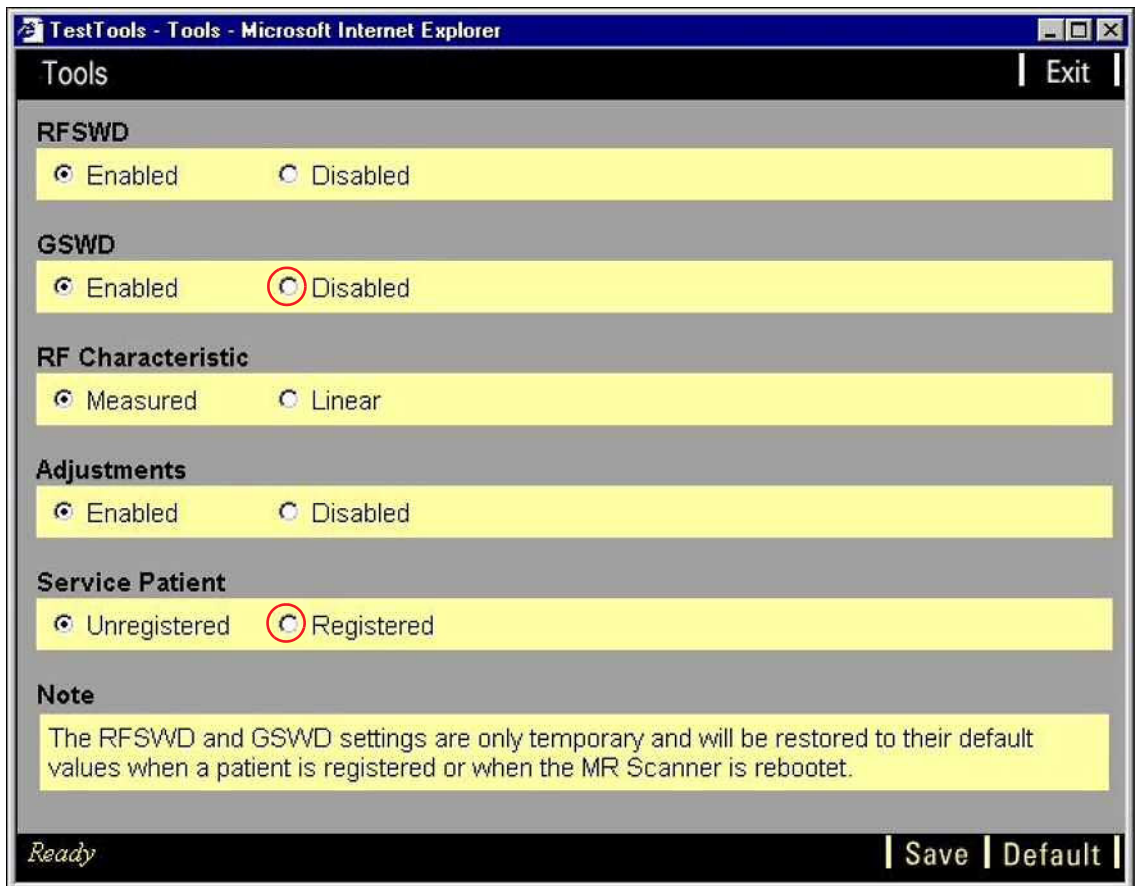
The tests described in this section are designed to make sure the gradient final stages of the Cascade Gradient Cabinet K2117 are performing as specified.

2.1.14.1 Gradient Risetime Check

- Position the Spherical Body Phantom and Loader.
- Open the "Local Service" platform.

- Select "TestTools / Tools".

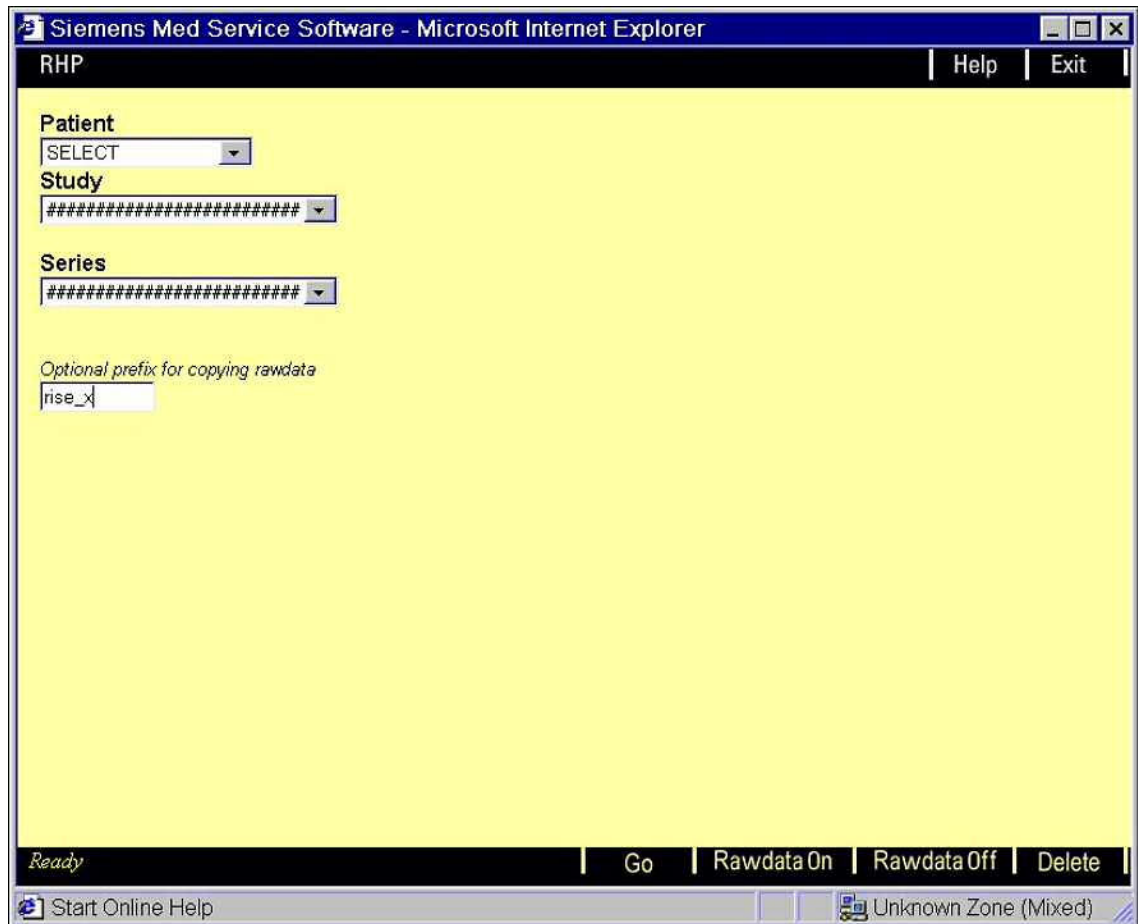
Fig. 20:



- Select "GSWD Disabled" and "Service Patient Registered" "Save" this selection and "Exit" the platform.

- Select "TestTools / RHP".

Fig. 21:



- Enter a filename for the raw data (e. g. rise_x)
- Open the "Examination" task card.
- Open the "Examination Explorer" and select the sequence SIEMENS - Sequence region - Service Sequences - Default Protocols - grad_ risetime.
- Copy the sequence into the examination queue (drag and drop).
- Select the orientation in the routine card according to the axis to be measured. The axis to be measured must be set as read out gradient
 - X axis: Orientation Transverse, phase encoding normal (A>>P)
 - Y axis: Orientation Transverse, phase encoding swapped (R>>L)
 - Z axis: Orientation Sagittal, phase encoding normal (A>>P)
- Open the "Contrast Card".
- The parameter TI(2) must be toggled to "1".

TI(2) -> "1"

- Keep the defaults for all other sequence parameters.
- Run the sequence. Click "Apply".

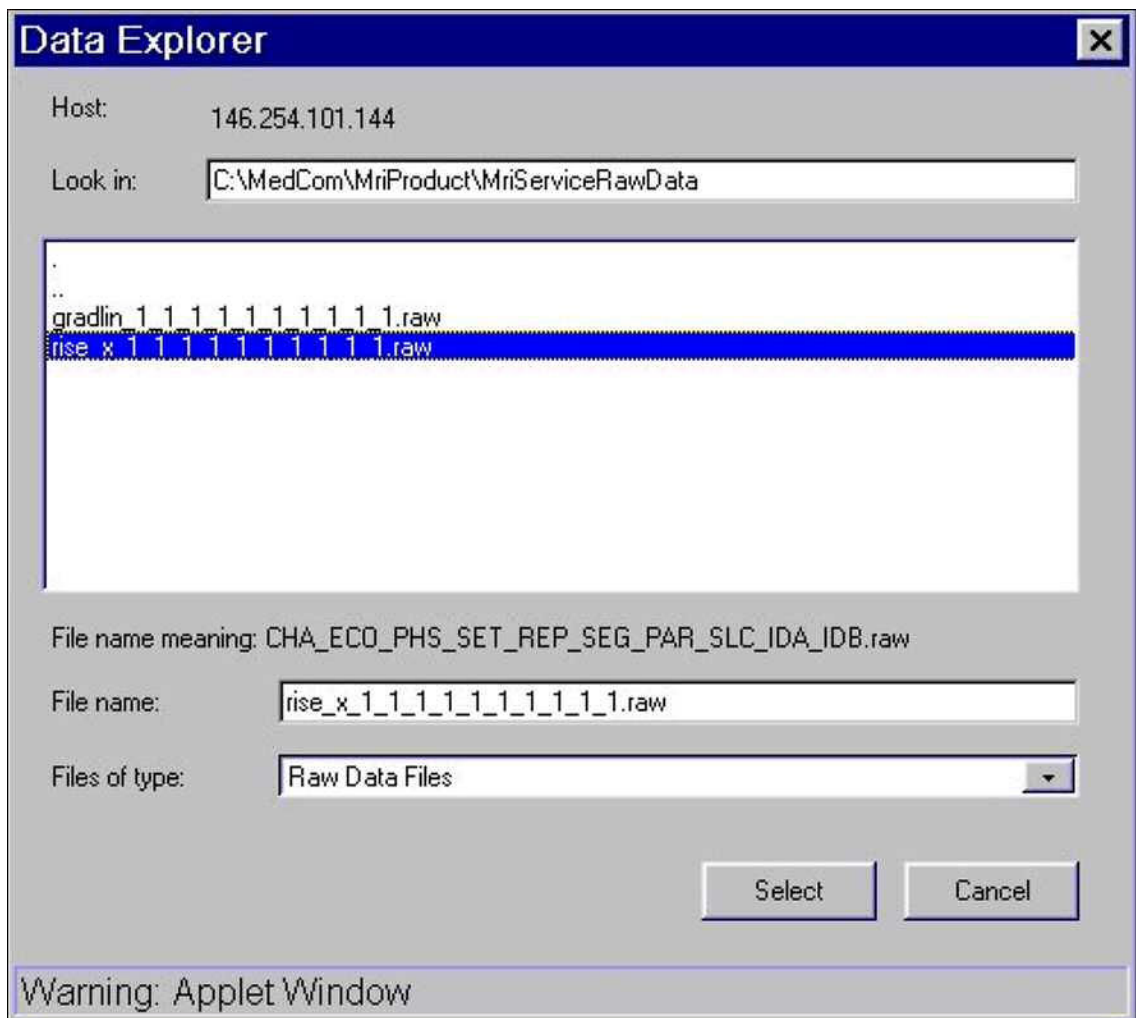
- RHP platform: Select "Go".

Fig. 22:



- RHP main menu: Select "Evaluate".
- Evaluation platform: Select "Browse".

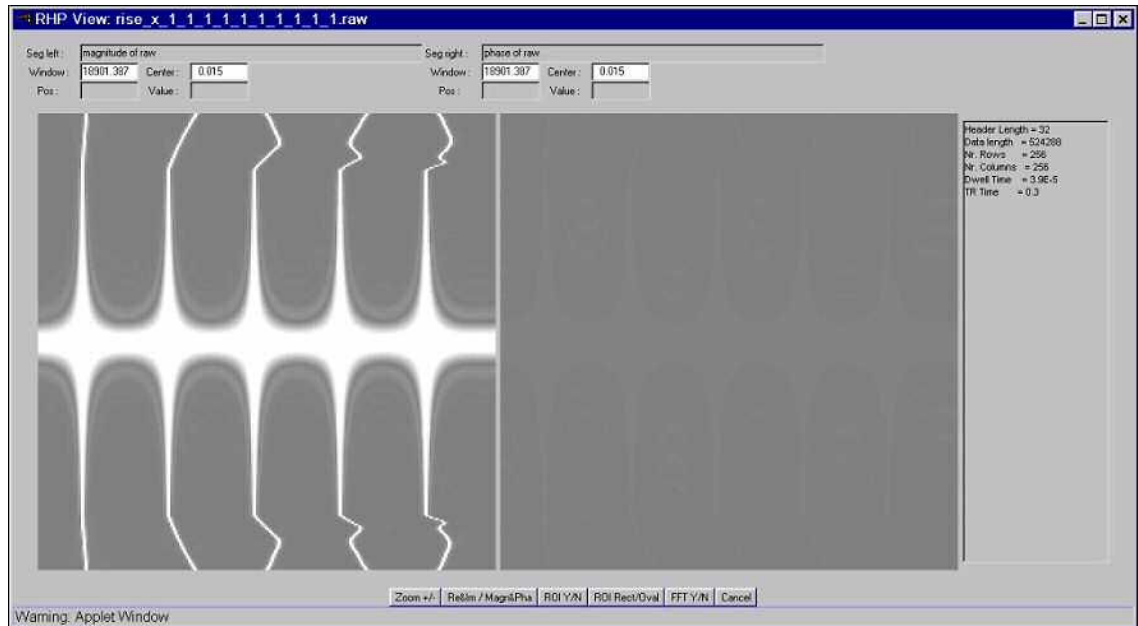
Fig. 23:



- Evaluation platform: Select the desired raw data file with doubleclick.

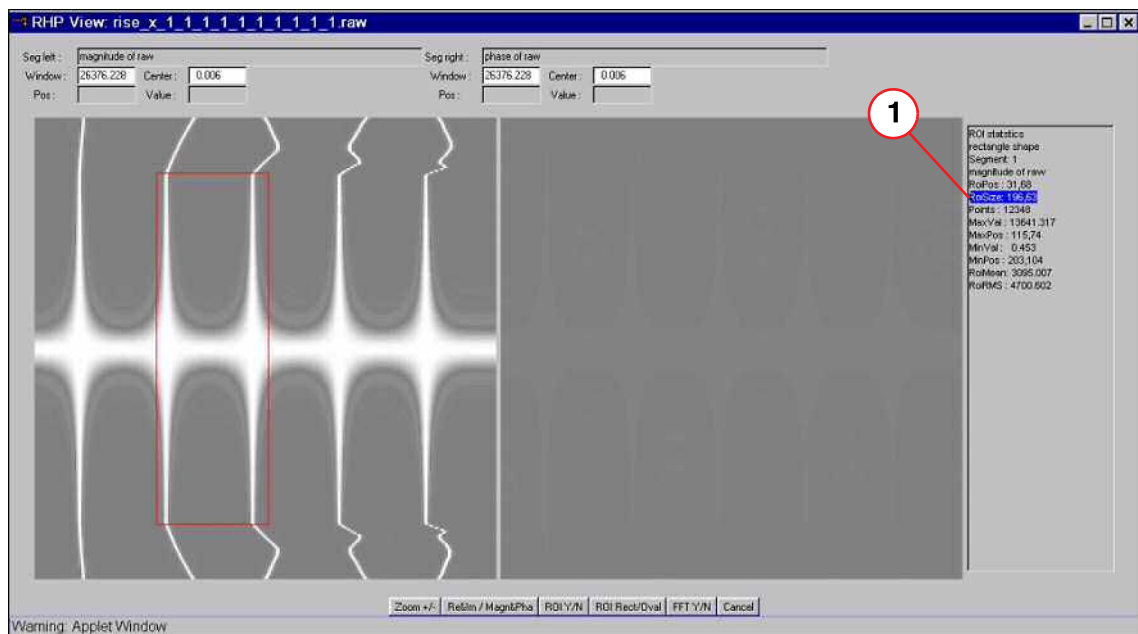
- Evaluation platform: Select "View"

Fig. 24:



- Viewing platform: Select "Zoom +/-", "Re&Im / Magn&Pha" and adjust the window and center to obtain the above representation.

Fig. 25:



(1) ROI Size (height, width)

- Click "ROI Y/N". Draw a rectangular ROI as shown above. The upper and lower boundaries of the ROI must be placed where the graphs of the gradient pulses start to deviate from the straight vertical lines. The width of the ROI is not important.
- Evaluate the value of ROI size (height):

ROI size ≥ 171

- Change the filename in the RHP platform for the measurement of the next axis (rise_y, rise_z).
- Open the grad_risetime protocol and change the orientation of the measurement.
- Repeat the measurement and evaluate it in the same way for the other orientations.

2.1.14.2 Possible reasons for this Test to fail:

- DC supply voltages for the final stages out of specification. Measure the incoming line voltage. Check tapping of the gradient transformer T1.
- Line resistance too high. (Refer to Gradient_Burst check.)
- UPS (if present) does not work correctly or switches into bypass mode.
- Hardware problems of GPA (final stages (partly) defective, one of the 5 individual DC supply voltages missing, problems of small signal unit GSSU).

2.1.15 GPA Fans Test Procedure

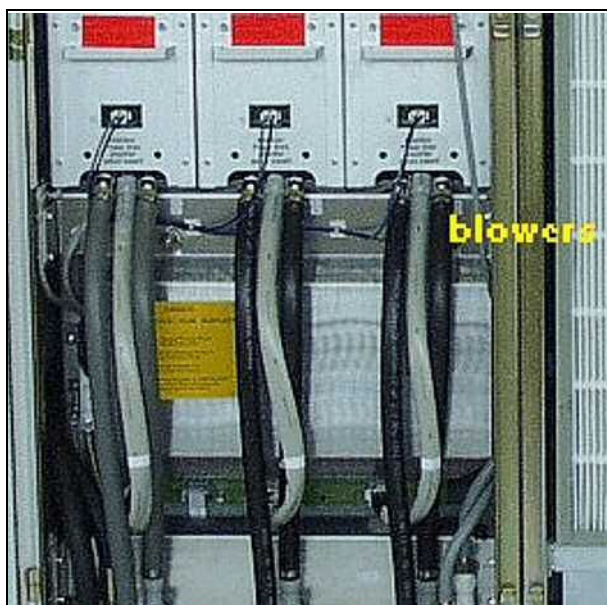
Check of fan function:

All five fans must work properly



The fan function of the first K2217 cabinets was controlled by temperature switches. With Service Instructions MR004/00/R, these cabinets had been modified to "Gradient amplifier blower- permanently on".

Fig. 26: Gradient Amplifier Blower



2.2 Gradient Supervision (VLM)

This document describes:

- How to troubleshoot VLM problems
- How to install and use the VLM diagnostic Software "Siemens Medical VSC"
- The mandatory checks after a VLM Fire Alarm before handing over the system to the customer



Useful additional VLM information can be found under:

CB-DOC>Installation>Installation_Instructions>Gradient_Supervision and
CB-DOC>Installation>Startup>System Start-up>Startup of the Gradient Supervision.

2.2.1 General Troubleshooting VLM

2.2.1.1 Safety Information



WARNING

High voltage

Contact with voltage-conducting parts will lead to death or serious physical injury.

- » **Adhere to the described steps for your own safety.**

2.2.1.2 Prerequisites

For individual troubleshooting steps, **additional items** may be required:

- Standard Windows PC (WIN7 is compatible)
- VLM diagnostic Software "Siemens Medical VSC" (to be downloaded from CS-Intranet MR-product page)
- RS232 cable 9p-SUB-D female/male (material number 10915402)
- only if PC has no RS232 port:
Adapter USB - RS232 (material number 10861204)
- Measurement device for ambient atmospheric pressure differences (to be sourced locally, e.g. supplier of air condition systems)
- T-Distributor to be additionally installed at VLM aspirating hose system (material number 10683922)
- Adapter_D22_D25 bushing for VLM hose connection (material number 10683929)
- Temperature indicators (clock stickers) for gradient cables (material number 7071145)

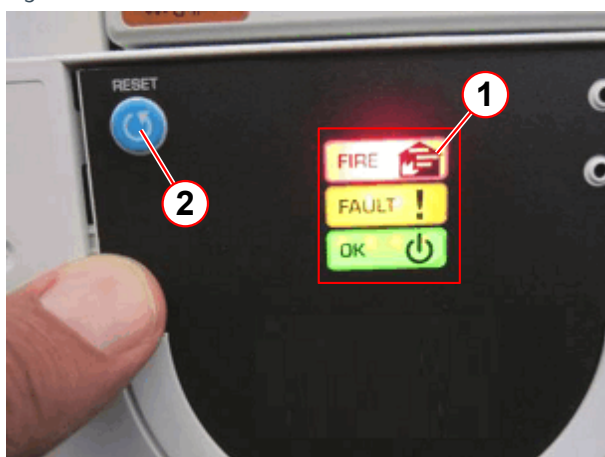
2.2.1.3 LEDs + Reset Button

Fig. 27: VLM front view



- The VLM-device has 3 status LEDs.
- A blue reset button is accessible when the front cover is opened.

Fig. 28: VLM LEDs + Reset Button



- (1) LEDs
(2) Reset Button

- GREEN = OK (default)
- YELLOW = FAULT (flashing, or static depending on the fault severity).
For SW versions >VB13, a customer pop-up is generated additionally.
- RED = FIRE (turns off the GPA modulator immediately and switches system to stand-by 1 min. later)
- The RESET-button is used for resetting FAULTs or normalizing the device (i.e., calibrating the 100% air flow value) .



The VLM device does only work properly under the following conditions:

- The air-flow through the VLM is between 12 and 24 l/min (allowed range for successful VLM normalization/calibration).
 - » Outside this range, the VLM device will show a warning (yellow FAULT LED on) and cannot be normalized/calibrated.
- The current air flow may deviate for max. +/- 30% from the normalized (calibrated) 100% value (RF-cabin door open/ closed).
 - » Deviations above this +/- 30% limit will lead to a FAULT condition of the VLM device and display of a customer pop-up.
 - » Deviations above +/- 20% from the normalized (calibrated) value will already lead to a "high air flow" fault but no customer alarm



At newer VLM devices with serial number $\geq$ B8712498, the maximum allowed deviation from the normalized (calibrated 100% air flow value) is $\pm 30\%$.
Older VLM devices ($<$ B8712498) have a smaller tolerance band of $\pm 20\%$.

2.2.2 Special Problem - Incorrect Air Flow

2.2.2.1 Problem Description + Troubleshooting Flowchart

The major problem currently seen in the field is an incorrect air flow through the VLM device.

There are several possible causes for that fault, see next section.



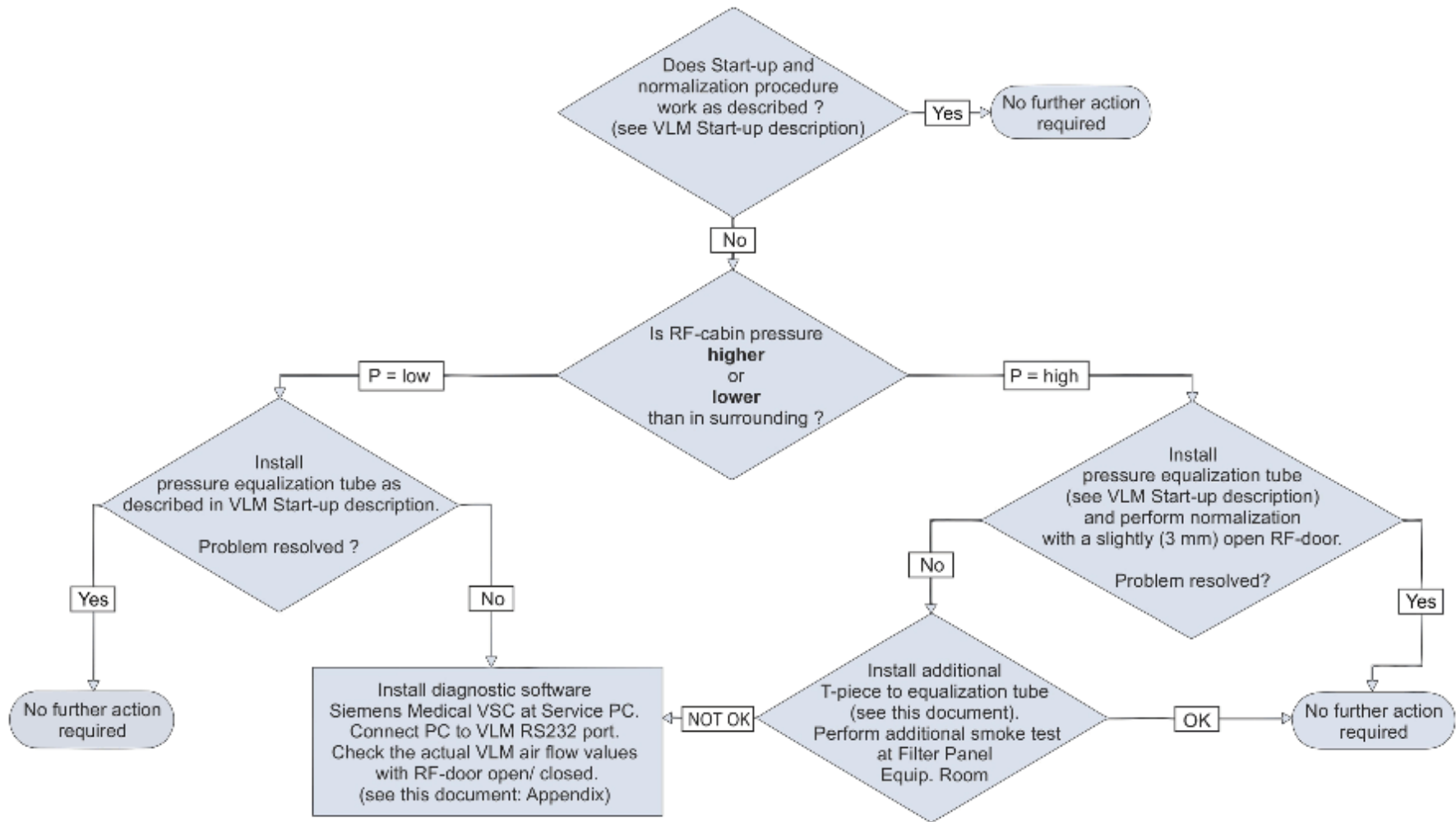
In case of incorrect air flow, the **VLM yellow FAULT LED is flashing or permanently ON.**

Fig. 29: Yellow Fault LED



Fig. 30: Flowchart - VLM air pressure problems

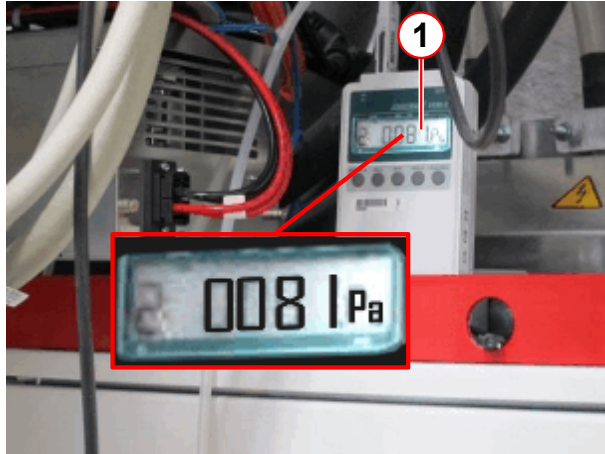
Air-pressure related VLM problems Troubleshooting Flowchart



2.2.2.2 Corrective Actions

Excessive Pressure Difference between RF-Cabin / Equipment Room

Fig. 31: Differential Pressure Measurement



(1) Example - Excessive diff. pressure > 81 Pa.

- The example on the left shows a site which was reporting VLM air flow problems.
- With help of a special measurement device (e.g. supplied by the air condition company), a differential pressure of 81 Pascal was measured between RF-cabin and equipment room!
- For a fault-free operation of the VLM, the required value is <50 Pa (0.5 mBar).
 - » The **optimum would be, to have NO differential pressure** at all!
 - » The air conditioning system has to be corrected (e.g., installation of additional exhaust air duct).

Fig. 32: Open RF door 3 mm



VLM normalization (calibration) with slightly open RF door:

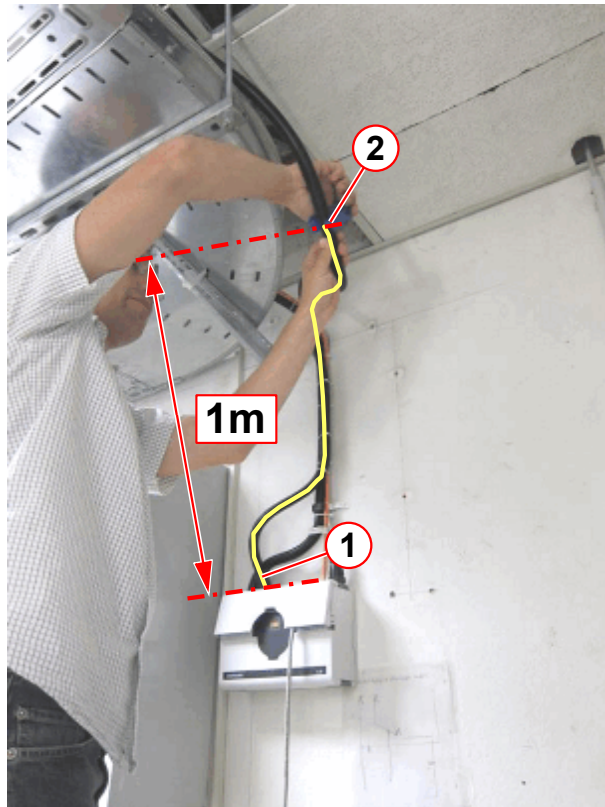
- A slightly modified VLM normalization procedure (calibration) of the 100% air flow value (with slightly open RF-door) can help to improve the situation. This is only applicable in case of a moderate pressure difference between RF-cabin and equipment room.
- The VLM device is powered up and operating for at least 90 sec. warm-up time.
- Open the RF cabin door slightly (~3 mm).
- Open the front cover of the device and push the RESET button for 10 sec. (→ Pos.2/Fig. 28 Page 46) until all LEDs are flashing.
- Release the RESET button and wait until the LEDs go off.
- When the LEDs start flashing again, press the RESET button once more to confirm the actual calibration.
 - » This results in a normalization at a mean cabin air pressure between maximum (door closed) and minimum (door open) and allows a maximum of tolerance from normalized (calibrated) value in both directions.



The diagnostic Software Siemens Medical VSC can be used to measure the VLM air flow and determine the optimum opening gap of the door for normalization.

See (→ Appendix - Diagnostic Software (Siemens Medical VSC) / Page 65).

Fig. 33: Cutting the Pressure Equalization Hose



- (1) VLM outlet
- (2) Cut hose 1 m after the VLM outlet

If the previous step did not help, install additional T-piece:

- An additional T-piece installed at the pressure equalization hose may help.
- Cut the pressure equalization hose about 1 m behind the VLM outlet (see left figure (2)).

Fig. 34: T-piece with 2 m hose



- Prepare a T-piece with approx. 2 m of hose as shown on the left.
- The T-piece can be ordered as a standard spare part with material number 10683922 (this T-piece has an additional reduction to 10 mm fitted, which must be removed).
- If needed, the black bushing at the VLM-in/ -outlet can be also ordered as a spare part (Adapter_D22_D25, material number 10683929).

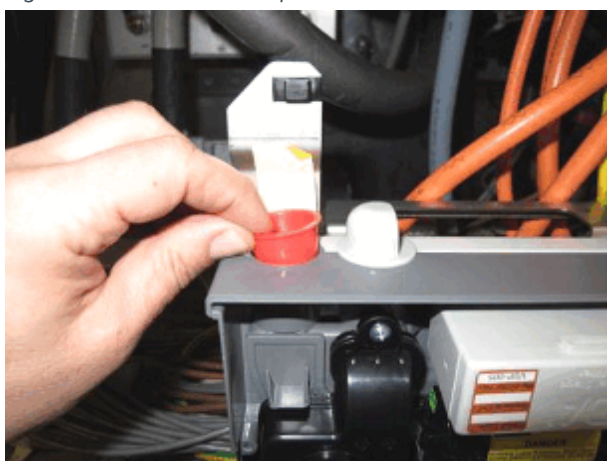
Fig. 35: T-piece connected



- Connect the T-piece to the previously cut ends of the pressure equalization hose as shown on the left.
- » The outlet of the 2 m hose must remain inside the equipment room.

VLM Blind Cap not removed

Fig. 36: Remove Blind Cap



- Here, a red blind cap was forgotten at the VLM input.
- This will result in zero flow and a FAULT condition.
- » Remove blind caps from the VLM before installing the aspirating hose system.

Leaking Hose Connections

Fig. 37: VLM Hose Connections

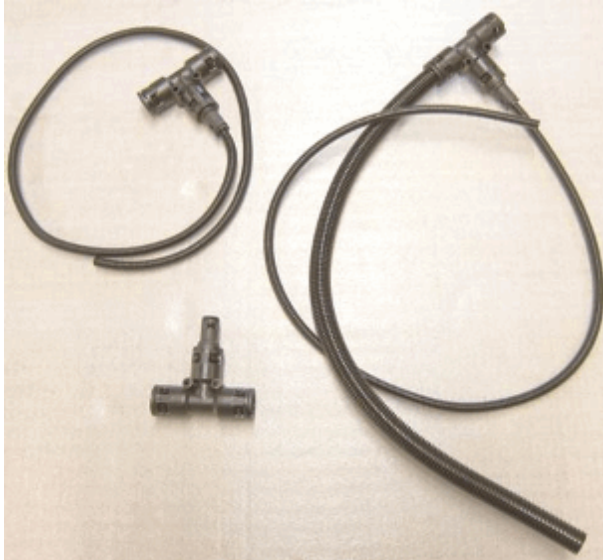


Fig. 38: Proper cut



Fig. 39: Improper cut

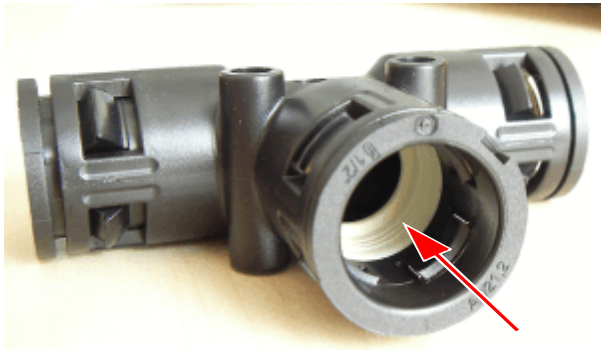


The aspirating hose system may leak in case:

- a tube is not completely pushed into it's manifold (T-piece...). Check turning moment between manifold and tube:
 - » Good fitting = noticeable resistance.
 - » Bad fitting = no resistance for turning the tube.
- a corrugated tube is not properly cut and does not have smooth edges.
 - » Cut the tube correctly with the delivered tool.
 - » See VLM Installation Instructions: **CB-DOC>Installation>Installation Instructions>Gradient Supervision**

Missing Joint Seals

Fig. 40: Joint Seal (white)

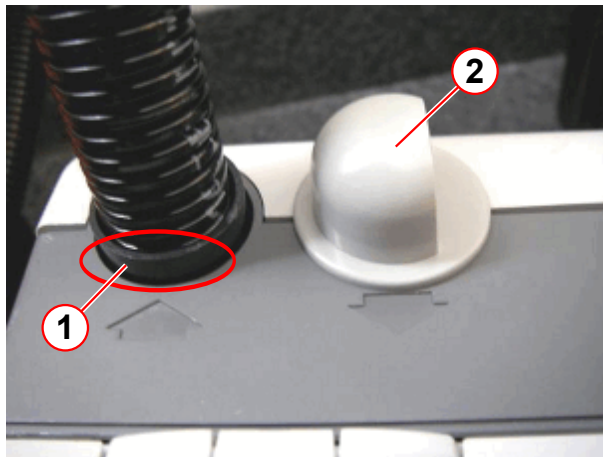


The white rubber joint seal must be properly fitted:

- In case of a FAULT condition, check the joints (T-pieces) for proper sealing.
- A missing seal easily results in an air flow difference of 1.4 l/min. (~ 7%).
 - » Check if the white rubber seal is fitted in each joint as shown on the left.
 - » In case the seal is missing, order a new T-piece (material number 10683922)

Missing Bushing

Fig. 41: VLM Hose Connection



- (1) Black Bushing
(2) Grey Outlet Cover

The black bushing must be always used when a hose is connected to the VLM inlet or outlet:

- If the black bushing (1) (Adapter_D22_D25, material number 10683929) is missing, the flow will increase dramatically for 9 l/min. (~ 43%).
- When the grey outlet cover (2) is removed, the flow increases for 0.4 l/min. (~ 2%)
 - » Check if the black bushing is fitted to the VLM inlet and, if applicable, to the VLM outlet hose connection.
 - » Check if the grey outlet cover is fitted to the VLM outlet (only in case no return hose is fitted).

2.2.3 Special Problem - Incorrect VLM Configuration

2.2.3.1 Problem Description



In case the VLM configuration jumper is set incorrectly, **the VLM will show NO fault indications, but the GPA will be permanently disabled with an error message.**

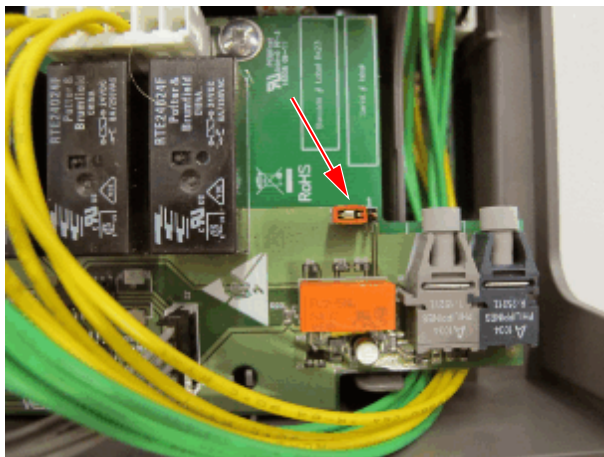
The VLM has a configuration jumper which must be set according to the system type.

- This jumper is inside the VLM..
- It defines the logic of the Modulator_off signal which is forwarded via a fiber-optic cable to the GSSU D70 I/O board, U1.
- In normal condition (NO fault), the light at FOC U1 must be ON.
- In case of a VLM FIRE Alarm (at Tim systems also in case of Gradient Coil overtemperature):
 - the light at U1 will be OFF
 - the GPA modulator will be switched off immediately
 - a GPA error message is generated.

2.2.3.2 Corrective Action

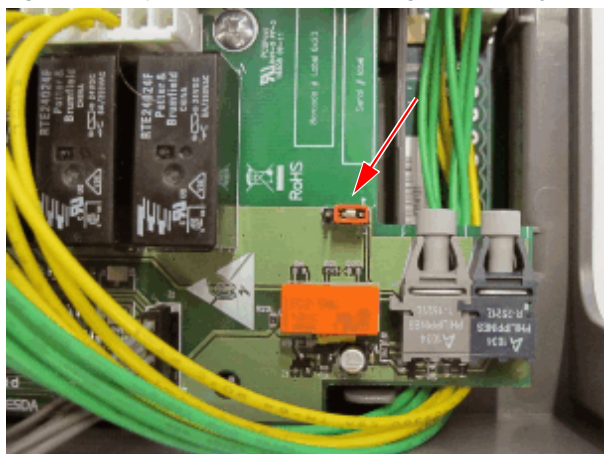
Check VLM Jumper Settings

Fig. 42: Jumper on 2-3 default setting - correct for Tim systems and Biograph mMR



- Open the VLM cover
- Inside, there is one jumper left of the fiber optic connectors:
 - » The default jumper setting is **2-3**, which is appropriate for all **Tim systems** and **Biograph_mMR**.
 - » On **Aera/ Skyra/ Prisma/ Avanto^{fit}/ Skyra^{fit} / Prisma^{fit}** systems, this jumper must be set to **1-2** as per (→ Fig. 43 Page 55)

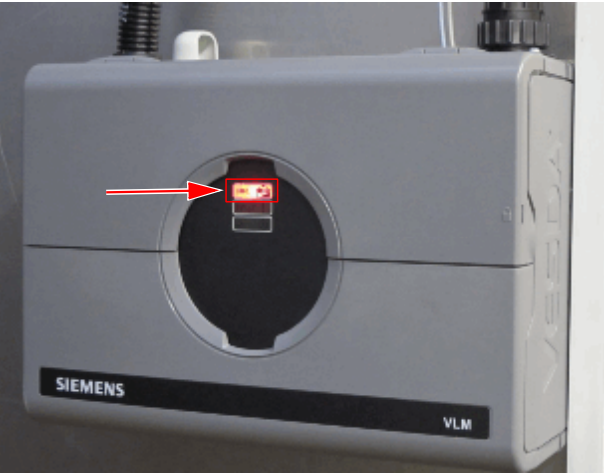
Fig. 43: Jumper on 1-2, correct setting for Aera/Skyra)



2.2.4 System Inspection after VLM FIRE Alarm

After a VLM FIRE Alarm was triggered for unknown reason, the following work steps must be successfully performed before handing over the system to the customer:

Fig. 44: VLM FIRE Alarm



2.2.4.1 Checklist



All system inspection steps must be successfully performed and checked with OK before handing over the system to the customer!

Tab. 1 System Inspection VLM FIRE Alarm - Checklist

All checks must be OK before returning the system back to the customer.			
Work Step	OK	Not OK	Procedure
MR system is powered-off WARNING: High voltage! See Safety Information.	☐	n.a.	(→ Prerequisites / Page 60)

All checks must be OK before returning the system back to the customer.			
Work Step	OK	Not OK	Procedure
NO signs of excessive heat around VLM Sample Points	☐	☐	(→ Check around Air Sample Points / Page 60)
Magnet - Lower Sample Plate:			
- Gradient Coax Cables			
- GC connections			
- Body Coil			
Magnet - Upper Sample Plate:			
- Gradient Coax Cables			
- Gradient Cables to RF Filter Panel			
- Gradient connections, bolts, nuts, washers			
RF-Filter Panel - Examination Room:			
- Gradient Cables			
- Bolts, nuts, washers			
- Gradient Filters			
RF-Filter Panel - equipment room:			
- Gradient Cables			
- Bolts, nuts, washers			
- Gradient Filters			
Inside EPC/ACC and GPA cabinet:			
- EPC/GPA Transformer			
- GPA Chokes			
- Other components			

All checks must be OK before returning the system back to the customer.			
Work Step	OK	Not OK	Procedure
NO External Causes found - NO craftsmen work in vicinity of site (welding, dust, etc.) - NO people smoking - NO use of fossil flammable material in the surrounding (heating systems) - NO air condition defects or distribution of external pollution through the air ducts (specially check inlet) - NO excessive environmental pollution	☐	☐	(→ Exclusion of External Causes / Page 60)
VLM Event Log - Download, analysis of Event Log - Event Log sent to RSC/ HSC	☐	☐	(→ Analysis of VLM Event Log / Page 61)
VLM Reset - FIRE Alarm + FAULT LED off	☐	☐	(→ Reset of VLM / Page 62)

All checks must be OK before returning the system back to the customer.			
Work Step	OK	Not OK	Procedure
Final Checks			(→ Final Checks / Page 62)
- System switched-off			
- NO activated temperature indicators (clock stickers at gradient cables) before system switch-on.			
WARNING: High voltage! See Safety Information.			
- System switch-on			
- NO increased smoke background level during running of service sequence "gradfree-pulse"			
- NO activated temperature indicators (clock stickers at gradient cables) after running "gradfreepulse"	☐	☐	
WARNING: High voltage! See Safety Information.			
- NO spikes found with QA>Spike Check			
- Check of Image Orientation QA>Image Orientation Check (only if gradient cables were removed, a new measurement must be performed)			

2.2.4.2 Prerequisites

**WARNING**

High voltage

Contact with voltage-conducting parts will lead to death or serious physical injury.

» **Adhere to the described steps for your own safety.**

- Ensure, the system is switched off
- Switch off circuit breakers:
 - for Aera/ Skyra/ Prisma/ Prisma^{fit}: **F21/EPC and F1/GPA**
 - for Avanto, Avanto^{fit}, Skyra^{fit}, Biograph_mMR, Verio and Trio A Tim System : **F2/GPA/GSSU and F1/GPA**
 - for Symphony A Tim System: **F6/GSSU in the ACC cabinet and F110/GPA at the mains box**
- Wait at least for 20 minutes to ensure the final stage capacitors are fully discharged.

2.2.4.3 Check around Air Sample Points

Check for any signs of excessive heat in the vicinity of all air sample points:

- At magnet lower sample plate (GC Coaxial Cable Connections)
- At magnet upper sample plate (Gradient Cable Connections, bolts, screws, etc.)
- At RF-filter plate Examination Room (Gradient Cable Connections, Gradient Filters)
- At RF-filter plate equipment room (Gradient Cable Connections, Gradient Filters)
- Inside EPC/ACC and GPA cabinet (System- and GPA-transformer, GPA-chokes, other electronic components)



Overheated components/ connections might be recognized by the following signs:

- Discoloration, color change (e.g. copper conductors)
- Melted insulation (cables)
- Black marks from sparks, smoke residuals
- Activated temperature indicators (clock stickers at gradient cables)
- Bad smell

2.2.4.4 Exclusion of External Causes

Exclude external causes (particle release) which might have triggered a VLM FIRE Alarm, e.g.:

- Craftsmen work (welding, dust, etc.)
- People smoking
- Use of fossil flammable material in the surrounding (heating systems)

- Air condition defects or distribution of external pollution through the air ducts (specially check inlet)
- Excessive environmental pollution



Check smoke graph in diagnostic Software "Siemens Medical VSC"
(→ Pos.2/Fig. 58 Page 69).

For countries with strong environmental pollution, a VLM-device with higher alarm threshold is available (material number 109 15 504, HSC release required).

2.2.4.5 Analysis of VLM Event Log

- Evaluate the VLM Event Log, see (→ Appendix - Diagnostic Software (Siemens Medical VSC) / Page 65).
- Save the VLM Event Log as file (→ VLM Event Log / Page 72) and send it to RSC/ HSC for further analysis.



Check time stamps of FIRE Alarm in the Event Log (→ VLM Event Log / Page 72).

If a FIRE Alarm was reported at times where the MR system was powered off, gradient connections and GPA are not the problem cause.

In this case, investigate External Causes (→ Exclusion of External Causes / Page 60) and within EPC/ACC cabinet.

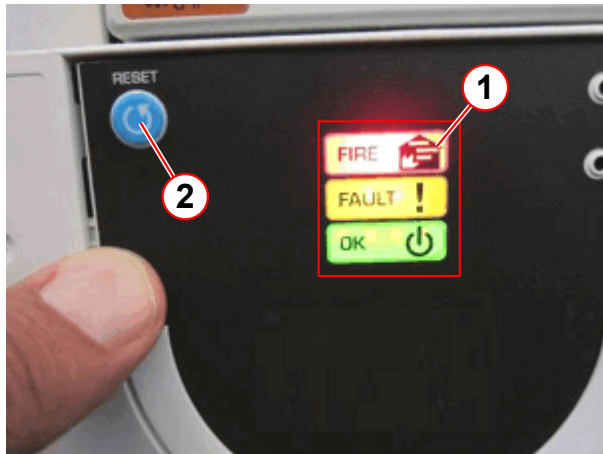
2.2.4.6 Reset of VLM

Fig. 45: VLM front view



- Open the VLM front cover.

Fig. 46: VLM LEDs + Reset Button



- (1) LEDs
(2) Reset Button

- Push the blue RESET-button (2).
 - » The red FIRE LED (1) should be extinguished.

2.2.4.7 Final Checks

- Prior to running measurements, check the condition of all temperature indicators (clock stickers) at the gradient cables.

CAUTION

High voltage!

Touching any of the gradient connections may result in serious injury or death.

- » Do not touch any of the gradient connections, or power down the system according to (→ Prerequisites / Page 60), prior to the following work steps.

Fig. 47: Temperature indicator (clock sticker) NOT activated

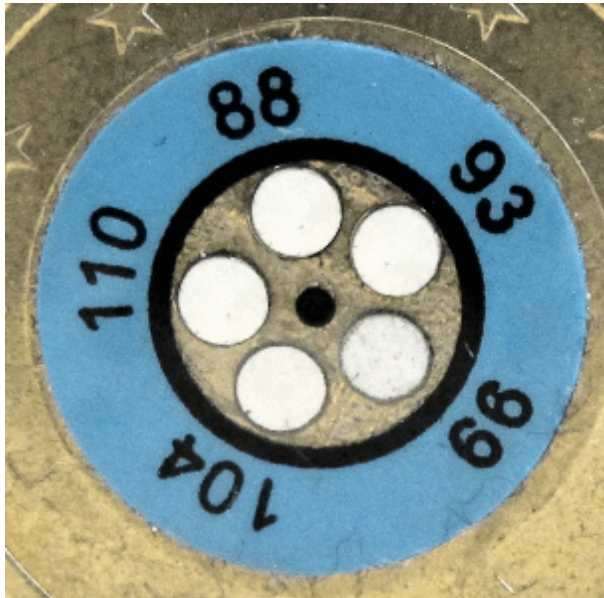
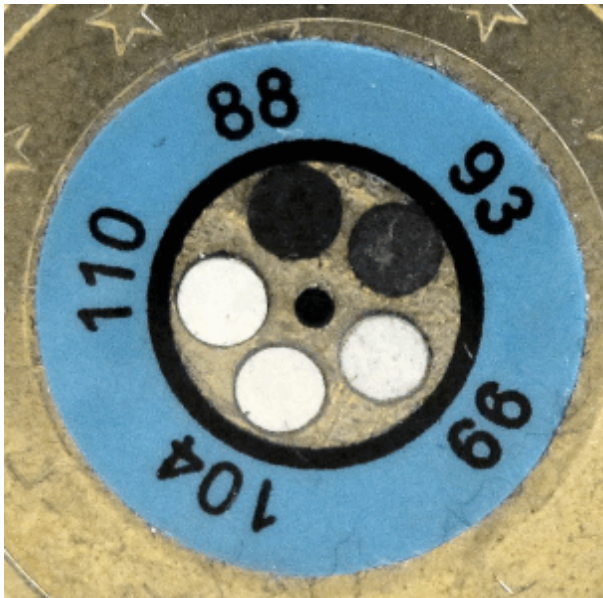


Fig. 48: Temperature indicator (clock sticker) activated



- Visually check all of the temperature indicators (clock stickers) for integrity.

- » **None of the labels must show a change in color!**

- An **activated temperature indicator** is a sign for a poor electric contact and high resistance due to loose or bad contacts.

- » Check/ correct the connection in doubt.
- » EXAMPLE - the left figure shows an activated temperature indicator, which has been exposed to $>93^{\circ}\text{C}$ and $<99^{\circ}\text{C}$.



In case of missing or damaged clock stickers, attach new temperature indicators (material number 7071145). Details see UI MR029/10/S.

- Switch on circuit breakers:
 - for Aera/ Skyra/ Prisma/ Prisma^{fit}: **F1/GPA and F21/EPC**
 - for Avanto, Avanto^{fit}, Skyra^{fit}, Biograph_mMR, Verio and Trio A Tim System : **F1/GPA and F2/GPA/GSSU**
 - for Symphony A Tim System: **First, F110/GPA at the mains box and then F6/GSSU in the ACC cabinet.** (Use this order to avoid damage!)
- Switch the MR system "ON" at the Alarm Box.

- Go to Service **Software>TestTools>Tools**.
- Register a Service Patient.
- Set adjustments to "Service Mode".
- Load service sequence "gradfreepulse".



This check is done to prove the integrity of the critical gradient connections at high load.

- Set the sequence parameters according to the following table:

Tab. 2 Sequence parameters to be set

Parameter	Value	Unit	Definitions
TI 1	10	mT/m	Gradient strength
TI 3	10	mT/m	Gradient strength
TI 5	10	mT/m	Gradient strength
TI 2	3	n.a.	
TI 4	3	n.a.	
TI 6	3	n.a.	
TI 7	1	n.a.	
TD 1	0.1	ms	Rise time
TD 3	0.1	ms	Rise time
TD 5	0.1	ms	Rise time
TD 2	0.3	ms	Flat top
TD 4	0.3	ms	Flat top
TD 6	0.3	ms	Flat top

- Set averages and number of measurements to achieve a time of ~28 min.
 - » Other parameters must not be changed.
- Run sequence while observing smoke graph with VLM diagnostic Software (→ Pos.2/Fig. 58 Page 69).
 - » The smoke background level must not increase during measurement!
Otherwise, check/ repair overheated gradient connections or other overheating parts (e.g. Body Coil).

- After measurements, check again the condition of the **temperature indicators**.



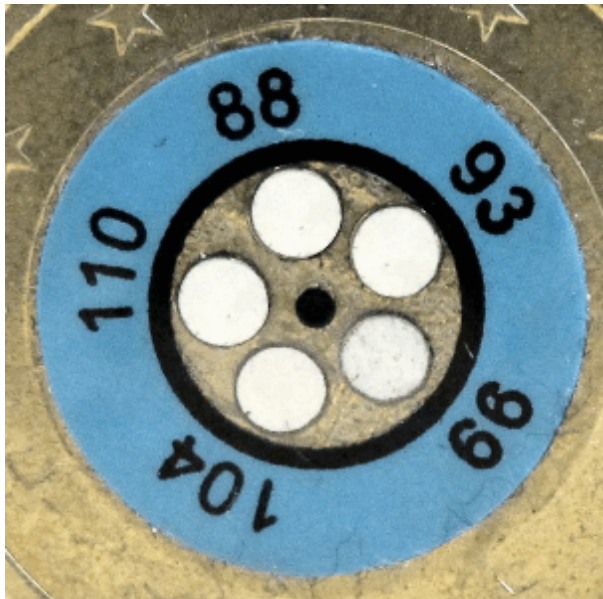
CAUTION

High voltage!

Touching any of the gradient connections may result in serious injury or death.

- » Do not touch any of the gradient connections, or power down the system according to (→ Prerequisites / Page 60), prior to the following work steps.

Fig. 49: Temperature indicator (clock sticker) NOT activated



- Visually check all of the temperature indicators (clock stickers) for integrity.
 - » None of the labels must show a change in color!

- Perform **QA>Spike Check** at least three times..
 - » If spikes were found, check the gradient connections again. If required, follow the **CB-DOC>TSG>TuneUp/QA>Spike Troubleshooting**
- If any of the gradient connections have been removed, **QA>Image Orientation Check** must be performed.

2.2.5 Appendix - Diagnostic Software (Siemens Medical VSC)

2.2.5.1 Overview

For a detailed troubleshooting of the VLM, the diagnostic Software "Medical VSC" is provided which can be installed at the PC.

2.2.5.2 Requirements

- Standard Windows PC (WIN7 is compatible)
- RS232 cable 9p-SUB-D female/male (material number 10915402)

- only if PC has no RS232 port:
Adapter USB - RS232 (material number 10861204)



In case your PC has no RS232 port:

- Use adapter USB - RS232 (material number 10861204). Other available standard adapters are usually not compatible with the VLM-device and no communication is possible.
- Installation of a driver might be required at the PC. Follow instructions printed at the packing of the USB - RS232 adapter.

2.2.5.3 Software Download

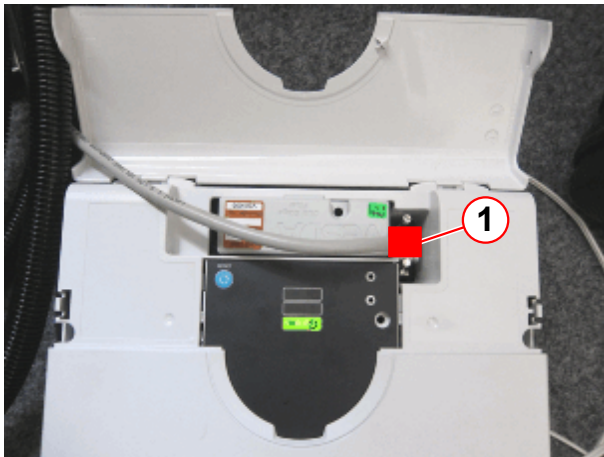
Fig. 50: Icon - Siemens Medical VSC



- Download the VLM Software_Win7.ZIP file from the Intranet **Product Information>MR>Downloads>Gradient_Supervision**
- Unzip it to your PC and install the SiemensMedicalSetup_30500.msi
- You will get an icon at your desktop:

2.2.5.4 Establishing Communication to VLM

Fig. 51: VLM - open top cover



(1) RS232 connector

- Connect the RS232 cable (male connector (1)) to the socket on front of the VLM-device (under the top cover).
- Connect the other end of the cable to your PC.

Fig. 52: Icon - Siemens Medical VSC



- At the PC, double-click the desktop icon:

- The VLM user interface will be shown (see below).

Fig. 53: VLM User Interface

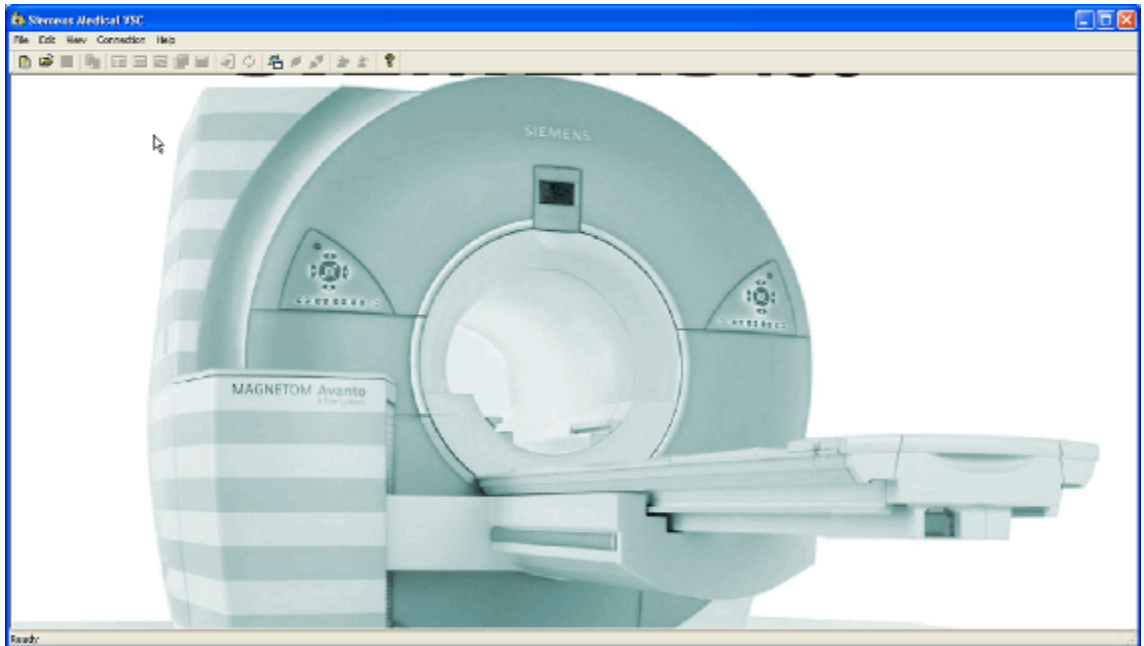
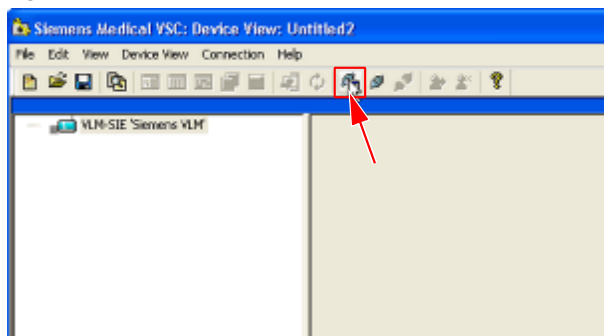
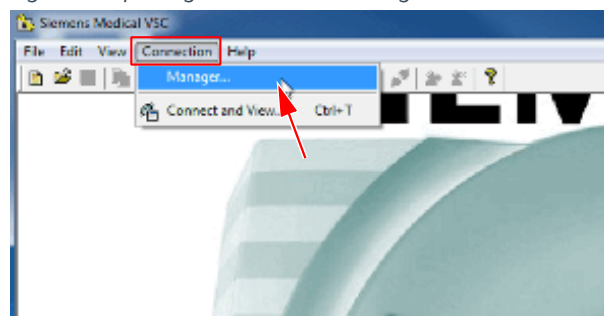


Fig. 54: "Connect" button



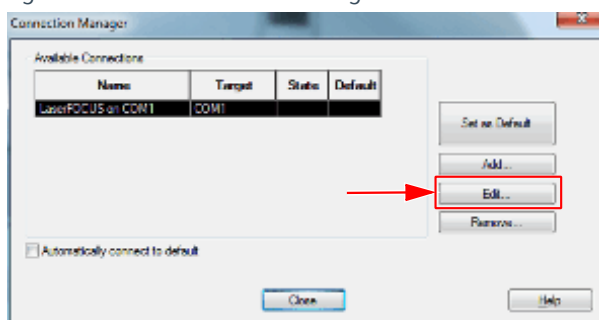
- Click the **"Connect"** button in order to start communication with the VLM-device.

Fig. 55: Opening "Connection Manager"



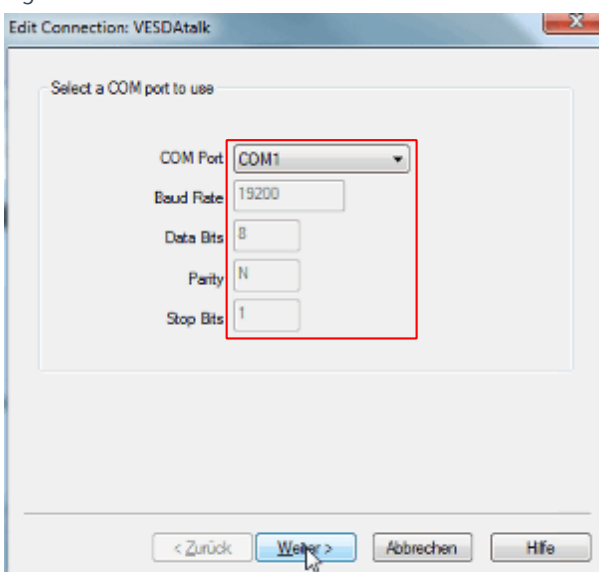
- If the communication parameters are not set properly, the communication will fail. In this case please open the Connection Manager.

Fig. 56: Edit - Connection Manager



- Click on “Edit”...

Fig. 57: Communication Parameters

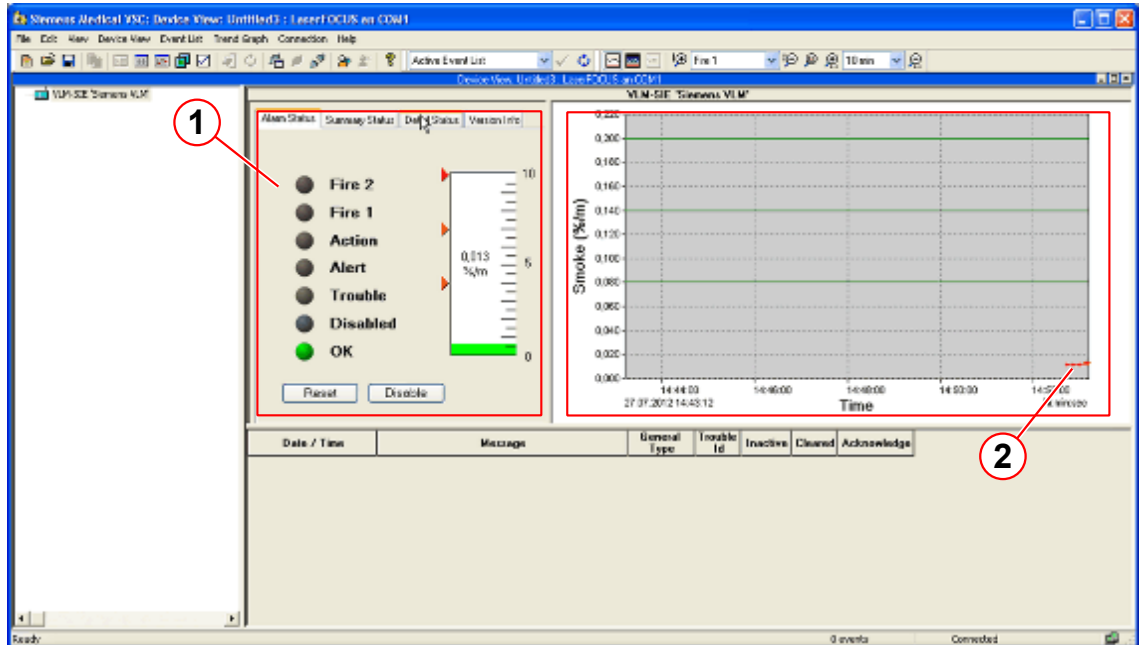


- ...and check/ correct the communication parameters according to the left figure.
 - COM Port used (here: COM1)
 - Baud Rate = 19200
 - Data Bits = 8
 - Parity = N
 - Stop Bits = 1

2.2.5.5 VLM User Interface

- After establishing communication to the VLM, the Device View will open with the Alarm Status (1) and Smoke Graph (2).

Fig. 58: Device View

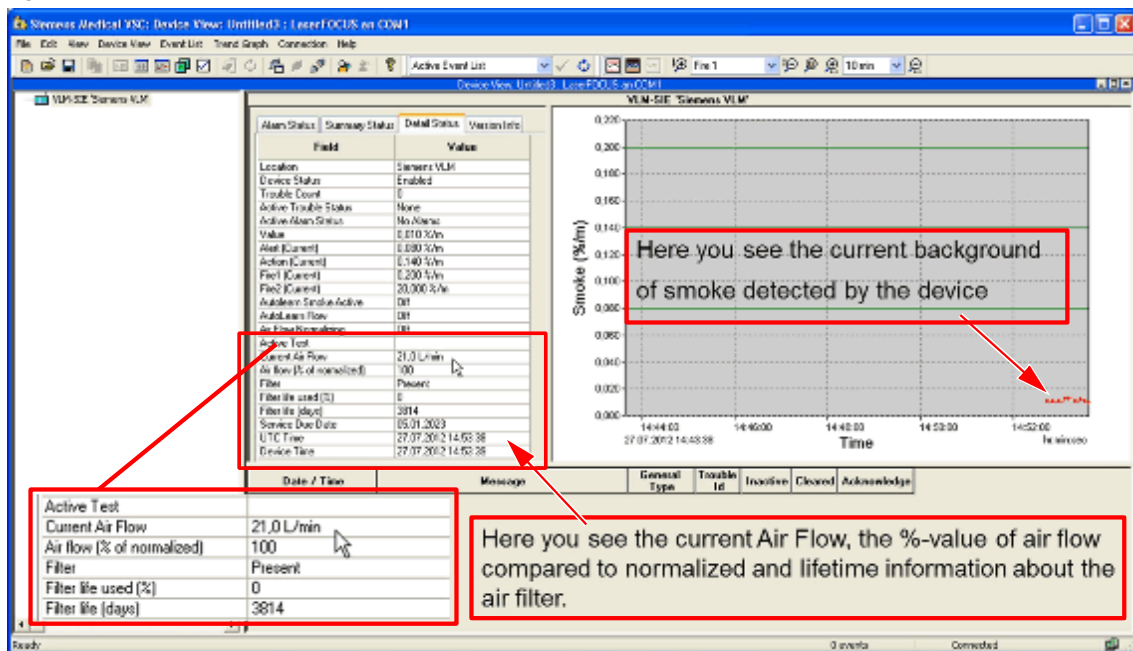


- (1) Alarm Status
- (2) Smoke Graph

2.2.5.6 Specified VLM Operating Range

- Click on the "Detail Status" Tab to see the most important information.

Fig. 59: "Detail Status" Tab



At newer VLM devices with serial number $\geq$ B8712498, the maximum allowed deviation from the normalized (calibrated 100% air flow value) is $\pm 30\%$.
Older VLM devices ($<$ B8712498) have a smaller tolerance band of $\pm 20\%$.

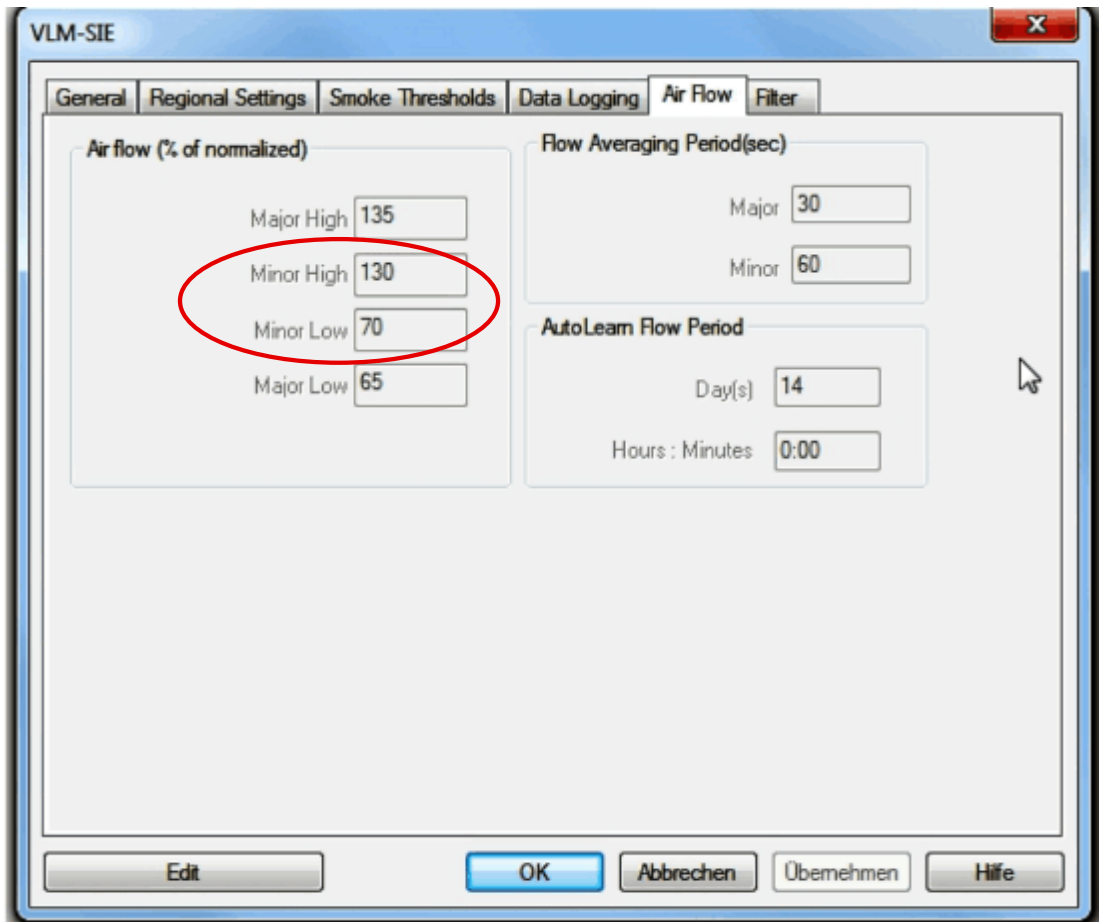


The VLM device does only work properly under the following conditions:

- The current air flow through the device must be in a range between 12 and 24 l/min (see figure above).
 - Outside this range, the VLM device will show a warning (yellow FAULT LED on) and cannot be normalized (calibrated).
- The current air flow may deviate for max. $\pm 30\%$ from the normalized (calibrated) 100% value if there is an atmospheric pressure difference >50 Pa (0.5 mBar) between RF-cabin and Equipment-room.
 - Deviations above this $\pm 30\%$ limit will lead to a FAULT condition of the VLM device and display of a customer pop-up. ($\geq$ B8712498)
 - Deviations above $\pm 20\%$ from the normalized (calibrated) value will already lead to a "high air flow" fault but no customer alarm.

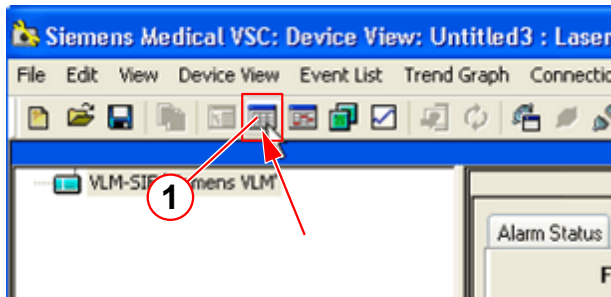
- Click the "EDIT" button, and select "EDIT" in the submenu to check the programmed air flow threshold.

Fig. 60: Air flow thresholds (VLM $\geq$ B8712498)



2.2.5.7 VLM Event Log

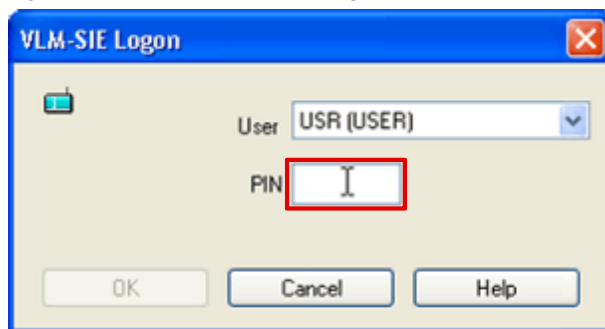
Fig. 61: Starting VLM Event Log



(1) Event Log button

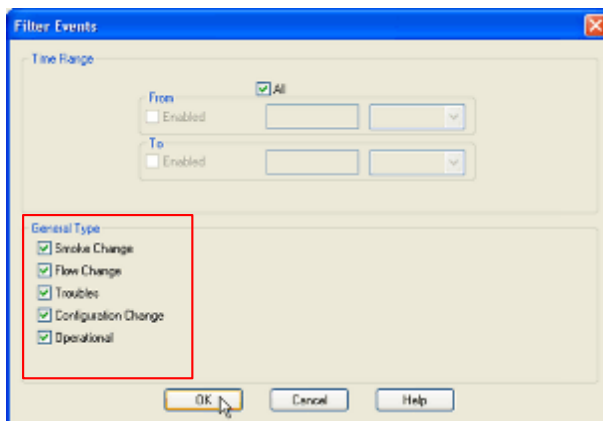
- Click on the “**Event Log**” button (1) to see the list of logged events.
- The list can be also downloaded with **File>Save as**.

Fig. 62: Password VLM Event Log



- The Software will ask you for a 4-digit PIN to get access to the Event Log.
- Enter **1111** and click OK.

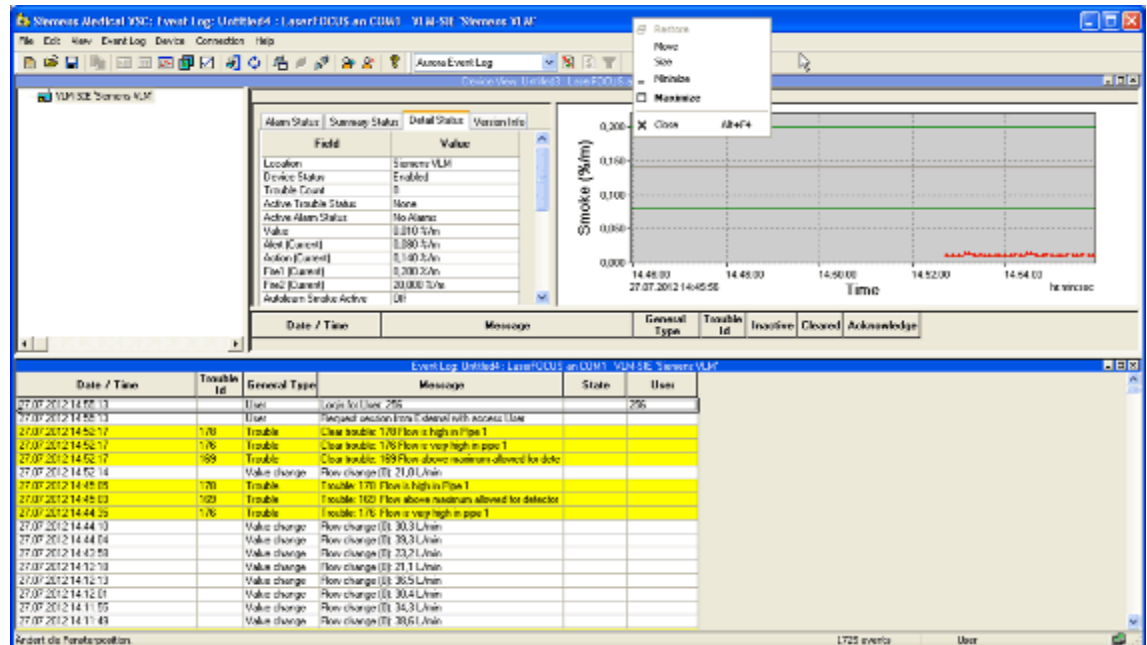
Fig. 63: Filter Settings



- By default, all filters are selected.
- Click “OK” to confirm configurations.

- Now you see the Event Log.
 - » Yellow marked events are FAULT conditions.
 - » Red events are FIRE conditions.
 - » The time stamps (VLM Device Time) might differ from the MR system time.

Fig. 64: Display of Events



There might be an offset between the Event Log time (= VLM Device Time) and the site-specific MR system time:

- The standalone battery-buffered **VLM clock cannot be set by the CSE.**
- The **VLM clock is set to UTC** (Universal Time, Coordinated) by default.
- Example: To correct the time stamp in the error log of the VLM device from UTC to e.g. CET (Central European Time), add +1 hr. Or, in case of daylight saving time CET DST, +2 hrs. must be added.

3.1 Test usage and interpretation of results

The tests are available in the Service software platform **TestTools**.

3.1.1 AMC - DAC gradient data test (DAC/Functional)

(Loop AMC/D5/X11 - GSSU/D11/X3 - AMC/D5/X11)

Test of 20 bit parallel data transmission and control signals via cable.

In case of errors, perform the AMC tests, check the cable connection between AMC and GPA. In addition, check the D11 K2217.

3.1.2 S_DAC - Nominal value (DAC/Functional)

(Loop GPA/GSSU/D139/X4 - AMC/D5/X15)

Nominal value for gradient current from DAC for x,y,z axes; stability and linearity measurement with several parameter variations, such as amplitude and timing.

If some of the test fail, check D11 K2217 (DAC), D139 K2217, cables to the AMC backplane. Perform the AMC tests.



If the test fails due to a regulation (loop) error, switch on the "service mode" (right switch S1 up on D4 or right switch S2 on D17). Now you can test the correct function of the DAC's.

3.1.3 S_CAV - Actual current value (X, Y, Z-axis/Functional)

(Loop GPA/GSSU/D12 - GPA/GSSU/D13)/X4 - AMC/D5/X15)

Actual value for gradient current from D12, actual value amplifier for x,y,z axes; stability and linearity measurement.

If some of the test fail, check D12 K2217 (regulator board) with "Regulator Check", cables to the AMC backplane. Perform the AMC tests.

4.1 Status of components K2217

Gradient Small Signal Unit Components	
D16 Service-CAN-board Normal Mode:	2 LED's "Trafo on" (V407,green) on; left LED (V403,green) is blinking (Loadware is ok, Application software (ASW) is running)
Standby:	2 LEDs (V407,green) off -> "Trafo off"; left LED (V403,green) is blinking (Loadware is ok, ASW is running)
D11 DAC	Switches S1/2 in down position
D12 Regulator Normal Mode:	LEDs V136/137 green on
Standby:	LEDs V136/137 green off
D14 Modulator Normal Mode:	green LED V59 on -> VCC +5V ok
Standby:	D14, LED "Modulator disabled" (V61/A2, red) is on in state 1,2 and 3.
GPA Small Power supply: Fan is running	
ACC: Circuit breaker F6 is switched on; Cutter switch F110 on top of ACC is switched on.	
Power stages: If the six green LED's behind the six holes in the front of the PS are on, the capacitors are charged.	

After "Power On" or Reset) all CAN units run through a self-test and start the firmware. The firmware checks whether the loadware has been loaded. It also checks the loadware code. The loadware is started if the consistency test is successful.

When the gradient processor is in the loadware mode, a logon command is sent every 500 milliseconds to the MPCU.

The gradient processor is the link between the gradient amplifier and the rest of the system. As a CAN unit, it is connected with the CAN bus to the system.

In loadware mode, the gradient processor is ready for operation with the exception of sending commands to the MPCU.

The gradient processor is fully operational after the MPCU has validated the version of the gradient processor via the login request and the software application has been started via a CAN command.

In other words, after the CAN unit has been booted and the loadware started, the application software is not fully available. However, after the login request by the MPCU, the application software is fully operational.

The application software of the gradient processor allows for the reliable operation of the overall component, namely the gradient amplifier. The most important tasks are:

- Setting the PI regulators for the three gradient axes
- Setting the different test loops for the service software (gradient stability/linearity tests)
- Service mode, switch on/off
- Setting the Power on/Standby mode

Service mode can be activated either by the service software or by setting manual (right) switch S1 on board D4 to the up position.

If service mode is active, the following errors will be handled as warnings, e.g. the modulators and regulators will not be disabled:

- K2217: Regulation error x,y,z (D12, LEDs /V7 yellow)
- K2217: No reaction of modulator

Version 01:

- The common print number (M6-015.840.01) from the Troubleshooting Guide is divided into separate Component print numbers.

Version 02:

- Link to "Gradient Error messages" has been added.

Version 03:

- Chapter (Procedures) - new section (→ Gradient Supervision (VLM) / Page 45) added.

Version 04:

- Chapter (→ Procedures / Page 16) - (→ Gradient Supervision (VLM) / Page 45) updated, air flow threshold +/- 30% up from serial number B8712498, Charm MR_00464686.

Version 05:

- Chapter (Procedures) - new section (→ Troubleshooting GPA Current Sensors / Page 26) added.

There are no Hazard IDs in this document.

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